The envelope property for unibranch projective varieties,
with an appendix on the singular Yau–Tian–Donaldson conjecture

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Abstract

We prove the envelope property for every real numerical class on an integral unibranch projective variety over an algebraically closed trivially valued field. Together with Boucksom–Jonsson's converse, this characterizes unibranchness by the plurisubharmonic envelope property. The Appendix applies the result to the uniform Yau–Tian–Donaldson conjecture for projective varieties with klt singularities.

1 Introduction

1.1 The envelope conjecture

Let k be algebraically closed with the trivial absolute value, and let X be an integral projective k-variety. Boucksom–Jonsson [BJ22] associate to each $\theta \in N^1(X)_{\mathbf{R}}$ a space $\mathrm{PSH}(X,\theta)$ of plurisubharmonic (psh) functions on X^{an}.

Continuity of envelopes and the variational solution of Monge–Ampère equations depend on the *envelope property*: for a fixed class θ one asks that for every nonempty family $(\varphi_i)_{i\in I} \subseteq \mathrm{PSH}(X,\theta)$ bounded from above, the upper semicontinuous (usc) regularization of the pointwise supremum,

$$\sup_i{}^{*}\,\varphi_i,$$

again lies in $\mathrm{PSH}(X,\theta)$. We abbreviate this statement as $\mathrm{EP}_X(\theta)$. It is the analogue of the classical fact that the usc regularization of the upper envelope of a locally uniformly bounded above family of psh functions on a domain in $\mathbf{C}^n$ is psh [Lel68].

Boucksom and Jonsson proved the envelope property for every numerical class on a smooth projective variety when $\mathrm{char}\, k = 0$ or $\dim X \leq 2$; see [BJ22, Theorem 5.20] for nef classes and [BJ24, Theorem A] in general. Conversely, if $\mathrm{EP}_X(\omega)$ holds for one ample class ω, then X is unibranch [BJ22, Theorem 5.12]. Here X is *unibranch* if every local ring $\mathcal{O}_{X,x}$ has one branch. This is equivalent to bijectivity of the normalization [Stacks, Tag 0C37 and Tag 0C3B], and, over the algebraically closed field k, to its being a universal homeomorphism (Lemma 8.1). This is also equivalent to local irreducibility of X^{an} [BJ22, §5.2]. Boucksom and Jonsson conjectured that unibranchness is sufficient for the envelope property in every class [BJ22, Conjecture 5.14].

The purpose of this paper is to prove the conjecture in full generality.

Theorem A. *Let X be an integral unibranch projective variety over an algebraically closed trivially valued field k, and let $\theta \in N^1(X)_{\mathbf{R}}$. Then every nonempty bounded-above family $(\varphi_i) \subseteq \mathrm{PSH}(X,\theta)$ satisfies*

$$\sup_i{}^{*}\,\varphi_i \in \mathrm{PSH}(X,\theta).$$

The theorem applies in every dimension and characteristic. Together with [BJ22, Theorem 5.12], it yields the following characterization.

Corollary 1.1. *For an integral projective variety X over k, the following are equivalent:*

(1) X is unibranch;

(2) $\mathrm{EP}_X(\theta)$ holds for every $\theta \in N^1(X)_{\mathbf{R}}$;

(3) $\mathrm{EP}_X(\omega)$ holds for some ample class ω.

Indeed (1) $\Rightarrow$ (2) is Theorem A, (2) $\Rightarrow$ (3) follows by choosing an ample class, and (3) $\Rightarrow$ (1) is [BJ22, Theorem 5.12].

1.2 Proof strategy

We first prove the envelope property for nef classes on smooth varieties (Theorem 3.1). In positive characteristic, asymptotic test ideals on the smooth product $X \times \mathbf{A}^1$ replace multiplier ideals on a resolution of a test configuration. A fixed generation twist gives Fubini–Study approximants in classes tending to the desired class, and a Frobenius-trace estimate identifies their decreasing limit with the envelope.

For a normal variety over a countable field, we use the Stein factorization $Y \to T \to X$ of a smooth alteration. The maximum principle of Gubler–Rabinoff [GR25, Theorem 6.1] makes the envelope on Y constant on the fibres of the birational map $Y \to T$. Local regularization then shows that the descended function is psh (Corollary 5.26), and weighted field norms descend the envelope property through the finite map $T \to X$ (Theorem 4.2). Flat base change of base ideals reduces a PL obstacle over an arbitrary field to a countable subfield (Theorem 6.1).

The local regularization argument follows Wu [Wu26]. On the open set where the modification is an isomorphism, regularizability follows directly from the global psh function upstairs. Polynomial hulls and norms along finite projections then extend it across the complement. This uses only the envelope property of projective space, already supplied by the smooth case.

The ample case yields homogeneous Siu division by first dividing with a fixed coefficient and then taking finite maxima over a base-point free linear system (Theorem 7.2). Homogeneous Legendre slices give the envelope property in arbitrary classes (Corollary 7.6). Finally, we descend through the normalization: Frobenius powers suffice in positive characteristic, while Roberts–Singh additive powers give a decreasing approximation after a fixed twist in characteristic zero (Theorem 8.3).

1.3 Related work

The smooth characteristic-zero case and the positive-characteristic surface case are due to Boucksom–Jonsson [BJ22; BJ24]. Wu [Wu26] developed a local theory of non-Archimedean psh functions and applied it to the envelope property of normal varieties in residue characteristic zero. Section 5 follows his approach; we indicate there where his arguments have to be corrected or extended to positive characteristic. Our positive-characteristic argument in the smooth case uses asymptotic test ideals [HY03; Mus13; BMS08], and the normalization argument uses subintegrality [Swa80; RS93]. Alterations [DJ96] and the maximum principle for classically psh functions [GR25] enter the passage from smooth to normal varieties.

1.4 Conventions

Throughout, k is an algebraically closed field endowed with the trivial absolute value. A *variety* is an integral separated scheme of finite type over k; unless stated otherwise, all varieties are projective. $N^1(X)_{\mathbf{R}}$ denotes the space of numerical classes of real Cartier divisors, and a *rational line bundle* is an element of $\mathrm{Pic}(X)_{\mathbf{Q}}$. Numerical classes are real unless the contrary is stated.

All nets are indexed by directed sets, and we pass freely to cofinal subnets. Countability of k is assumed only in Section 5 and in Step 1 of the proof of Theorem 6.1.

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2 Preliminaries

Let X be an integral projective variety over k. We fix notation for its analytification and psh functions, and record the behavior of Fubini–Study classes under finite morphisms.

2.1 The analytification and its semivaluations

We regard k as a trivially valued non-Archimedean field and write X^{an} for the Berkovich analytification of X [Ber90]; see [BJ22, §1] for a detailed discussion of the trivially valued case. A point $v \in X^{\mathrm{an}}$ is a *semivaluation*: it is given by a point $\xi \in X$, the *support* of v, together with a real valuation of the residue field $\kappa(\xi)$, trivial on k. We use additive notation, so $v(f) \in \mathbf{R} \cup \{+\infty\}$ for a local function f, with $v(f) = +\infty$ exactly when f vanishes along the support. Since X is proper, every $v \in X^{\mathrm{an}}$ admits a unique *center* $c(v) \in X$, the point such that $v \geq 0$ on $\mathcal{O}_{X,c(v)}$ and $v > 0$ on the maximal ideal.

The analytification is a compact Hausdorff topological space, functorial in X; for a morphism $f\colon X \to X'$ we write f^{an} for the induced map, and we normally abbreviate $f^{\mathrm{an}}(v)$ to $f(v)$.

Two classes of points play a special role. The *trivial valuation* v_{triv} is the zero valuation on $k(X)^\times$. The set $X^{\mathrm{div}} \subseteq X^{\mathrm{an}}$ of *divisorial points* consists of v_{triv} together with all positive rational multiples $c\,\mathrm{ord}_E$, where E is a prime divisor on some normal projective birational model of X and $c \in \mathbf{Q}_{>0}$. Points of X^{div} have support the generic point of X; the support of a general point of X^{an} may be any point of X.

The multiplicative group $\mathbf{R}_{>0}$ acts continuously on X^{an} by scaling: $t \cdot v$ is the semivaluation tv with the same support. The induced *scaling action* on functions is

$$(t \cdot u)(v) \coloneqq t\, u(t^{-1}v), \qquad t > 0, \tag{2.1}$$

and a function u is *homogeneous* if $t \cdot u = u$ for all $t > 0$.

Finally, let $S = X \times \mathbf{A}^1$ be the trivial test configuration, with coordinate t on $\mathbf{A}^1$. Every $v \in X^{\mathrm{an}}$ has a *Gauss extension* $\sigma(v)$: the unique $\mathbf{G}_m$-invariant semivaluation on S whose restriction to functions from X is v, normalized by $\sigma(v)(t) = 1$; explicitly, $\sigma(v)(\sum_\lambda s_\lambda t^\lambda) = \min_\lambda\{v(s_\lambda)+\lambda\}$ for local sections s_λ of $\mathcal{O}_X$. By [BJ22, §1.3], σ is a bijection from X^{an} (resp. X^{div}) onto the set of $\mathbf{G}_m$-invariant semivaluations (resp. divisorial valuations) w on S with $w(t) = 1$; in particular $\sigma(X^{\mathrm{div}}) \subseteq S^{\mathrm{div}}$. For a nonzero (equivariant) coherent ideal $I \subseteq \mathcal{O}_S$ we set

$$\phi_I(v) \coloneqq -\sigma(v)(I), \qquad v \in X^{\mathrm{an}},$$

where $\sigma(v)(I)$ is the minimum of $\sigma(v)$ over local generators of I at the center. We explicitly allow $\phi_I(v) = -\infty$, which happens when the support of $\sigma(v)$ lies inside the zero locus of I.

The support of $\sigma(v)$ is the generic point of $\operatorname{supp}(v) \times \mathbf{A}^1$, which does not lie on the central fibre $X \times \{0\}$. Hence, if $\mu\colon \mathcal{X} \to S$ is a projective birational morphism that is an isomorphism outside the central fibre, $\sigma(v)$ lifts uniquely to a semivaluation on $\mathcal{X}$, and we define ϕ_I in the same way for ideals $I \subseteq \mathcal{O}_{\mathcal{X}}$. For a Cartier divisor D on $\mathcal{X}$ supported on the central fibre we put $\phi_D(v) \coloneqq \sigma(v)(D)$, the value of $\sigma(v)$ on a local equation of D; thus $\phi_{\mathcal{O}_{\mathcal{X}}(-D)} = -\phi_D$ for $D \geq 0$.

2.2 Plurisubharmonic functions and the envelope property

For a rational line bundle $L \in \operatorname{Pic}(X)_{\mathbf{Q}}$, a *rational, generically finite Fubini–Study function* for L, or simply a *Fubini–Study function*, is a function of the form

$$\varphi = \frac{1}{m} \max_j \{\log |s_j| + \lambda_j\}, \tag{2.2}$$

where $m \in \mathbf{Z}_{>0}$ is such that mL is an honest line bundle, the $s_j \in H^0(X, mL)$ form a finite family of sections, not all zero, and $\lambda_j \in \mathbf{Q}$. Here $\log |s|(v) = -v(s)$ is computed in a local trivialization at the center of v; the choice of trivialization does not matter. We denote by $\mathcal{H}^{\mathrm{gf}}_{\mathbf{Q}}(X, L)$, or $\mathcal{H}^{\mathrm{gf}}_{\mathbf{Q}}(L)$ when X is understood, the set of such functions; this is the space denoted $\mathcal{H}^{\mathrm{gf}}_{\mathbf{Q}}(L)$ in [BJ22; BJ24].

When applying [BJ22, Definition 4.1], we also allow real weights in (2.2); the notation $\mathcal{H}^{\mathrm{gf}}_{\mathbf{Q}}$ remains reserved for rational weights. This gives the same psh class. Indeed, for each member f_j of a decreasing real-weight net and each $\epsilon > 0$, round the weights upwards to obtain a rational-weight function r with $f_j + \epsilon \leq r \leq f_j + 2\epsilon$. Order the choices by increasing j, decreasing r and decreasing ϵ. They form a directed set: after taking a common later original index, choose twice the new ϵ smaller than both old ones. The inequalities are preserved even at common zeros, since rounding does not change a pole set. The resulting rational-weight net has the same limit and class limit.

Since the sections may have common zeros, a Fubini–Study function may take the value $-\infty$; it is finite at v_{triv}, and more generally at every point whose support is not contained in the common zero locus. We write $\mathcal{H}_{\mathrm{hom}}(X, L) \subseteq \mathcal{H}^{\mathrm{gf}}_{\mathbf{Q}}(X, L)$ for the subset of *homogeneous* Fubini–Study functions, namely those of the form (2.2) with all $\lambda_j = 0$.

For $\theta \in N^1(X)_{\mathbf{R}}$, Boucksom–Jonsson define the class $\mathrm{PSH}(X, \theta)$ of θ-psh functions $u\colon X^{\mathrm{an}} \to \mathbf{R} \cup \{-\infty\}$, $u \not\equiv -\infty$; when $\theta = c_1(L)$ we also write $\mathrm{PSH}(X, L)$. We further write

$$\mathrm{PSH}_{\mathrm{hom}}(X, \theta) \coloneqq \{u \in \mathrm{PSH}(X, \theta) : u \text{ is homogeneous}\}.$$

We use the following properties of $\mathrm{PSH}(X, \theta)$. Recall that sup* denotes the upper semicontinuous (usc) regularization of a supremum.

Proposition 2.1. *Let X be an integral projective variety and $\theta \in N^1(X)_{\mathbf{R}}$.*

(1) If $\theta_i \to \theta$ and $u_i \in \mathrm{PSH}(X, \theta_i)$ decreases pointwise, its limit belongs to $\mathrm{PSH}(X, \theta)$ unless it is identically $-\infty$, provided that θ is nef or that all u_i are homogeneous. The assertion also holds for all classes when X is unibranch, as proved after the main theorem in Corollary 8.4. We use PSH2 *only in the nef case; the homogeneous case is proved in Lemma 2.3.*

(2) Every θ-psh function is usc and real valued on X^{div}. If $u \in \mathrm{PSH}(X, \theta)$ and $h\colon X^{\mathrm{an}} \to \mathbf{R} \cup \{-\infty\}$ is usc, then $u \leq h$ on X^{an} if and only if $u \leq h$ on X^{div}. In particular, every θ-psh function is determined by its restriction to X^{div}.

(3) *Every* $\varphi \in \mathcal{H}^{\mathrm{gf}}_{\mathbf{Q}}(X, L)$ *belongs to* $\mathrm{PSH}(X, c_1(L))$. *Conversely, every* $u \in \mathrm{PSH}(X, \theta)$ *is the pointwise limit of a decreasing sequence* $\varphi_i \in \mathcal{H}^{\mathrm{gf}}_{\mathbf{Q}}(X, L_i)$ *with rational classes* $c_1(L_i) \to \theta$*; if* u *is homogeneous, the* φ_i *can be taken in* $\mathcal{H}_{\mathrm{hom}}(X, L_i)$.

(4) *If* $f\colon X' \to X$ *is a surjective morphism of integral projective varieties, then* f^{an} *is surjective, and* $f^* := (\cdot) \circ f^{\mathrm{an}}$ *maps* $\mathrm{PSH}(X, \theta)$ *into* $\mathrm{PSH}(X', f^*\theta)$.

(5) *Finite maxima of* θ*-psh functions are* θ*-psh, and so is* $u + c$ *for* $u \in \mathrm{PSH}(X, \theta)$ *and* $c \in \mathbf{R}$. *Ordinary scalar multiplication satisfies* $au \in \mathrm{PSH}(X, a\theta)$ *for* $a > 0$, *and* $\mathrm{PSH}(X, \theta) + \mathrm{PSH}(X, \eta) \subseteq \mathrm{PSH}(X, \theta + \eta)$. *If* $\eta - \theta$ *is nef, then* $\mathrm{PSH}(X, \theta) \subseteq \mathrm{PSH}(X, \eta)$*; this follows by adding the zero function in the nef class* $\eta - \theta$.

(6) *Every* $u \in \mathrm{PSH}(X, \theta)$ *attains its supremum at the trivial valuation:* $\sup_{X^{\mathrm{an}}} u = u(v_{\mathrm{triv}}) \in \mathbf{R}$. *Every homogeneous* θ*-psh function vanishes at* v_{triv}*, hence is nonpositive.*

For the nef case of (1), choose an ample class ω and $\delta_i \to 0$ so that $\eta_i = \theta + \delta_i\omega$ is ample and $\eta_i - \theta_i$ is ample, after refining the indexing if necessary. Promote u_i to $\mathrm{PSH}(X, \eta_i)$ by (5). In an ample class, truncating Fubini–Study approximants by constants gives decreasing approximations by finite-valued continuous Fubini–Study functions in classes tending to η_i; use generating sections of ample multiples for the constant terms. For these finite-valued families the compactness diagonalization in [BJ22, Lemma 4.6] is valid: adding a smaller positive constant gives a strict comparison with each old, finite-valued approximant on the compact space X^{an}. Thus they form one decreasing Fubini–Study net with class limit θ. This proves (1) in the nef case directly from the definition. The singular homogeneous case is treated below.

In (2), upper semicontinuity is [BJ22, Proposition 4.12(i)], finiteness on X^{div} is [BJ22, Proposition 1.28 and Corollary 4.17], and the comparison is [BJ22, Theorem 4.22].

In (3), the first assertion is part of the definition of psh functions [BJ22, Definition 4.1], and sequential regularization is [BJ22, Corollary 12.18]: its sequence-extraction argument, [BJ22, Theorem 12.16], takes place after promotion to a fixed ample class. Homogeneous regularization can be obtained without general decreasing stability. For a rational-weight Fubini–Study approximant $f \geq u$ to a homogeneous u, apply the homogeneous projection $Hf = \inf_{t>0} t \cdot f$. The rational-polytope monomial formula in the proof of [BJ22, Theorem 6.15] makes Hf homogeneous Fubini–Study in the same line bundle. Indeed, for $f = m^{-1} \max_l(\log|s_l| + \lambda_l)$ the relevant polytope is $\{w_l \geq 0 : \sum_l w_l = 1,\ \sum_l w_l\lambda_l \geq 0\}$; clearing the denominators of its vertices gives products of the s_l. Vanished factors are treated by restricting to the face supported on the nonvanishing sections, with zero exponents omitted. The operator H commutes with decreasing infima, so these projections decrease to u.

Items (4) and (5) can also be checked directly on defining approximants, avoiding general decreasing stability in [BJ22, Theorem 4.7]. Pull back the sections for (4); for sums use products of sections, and for finite maxima first promote finitely many approximating classes to one nearby rational line bundle by tensoring with globally generated ample differences. Real translates follow by rounding constants from above. For a real scalar $a > 0$, choose rational $a_m \uparrow a$ and an upper bound C for a tail of a decreasing regularization f_j of u. Then $a_m(f_j - C) + aC$ is a decreasing Fubini–Study net with limit au and classes tending to $a\theta$. Finally, zero belongs to every nef class by approximation by globally generated ample rational classes, which gives the class inclusion in (5). These arguments recover the properties cited in [BJ22, Theorem 4.7 and Equation (4.3)]; in (4), the alternative $f^*u \equiv -\infty$ of [BJ22, Theorem 4.7(iii)] is excluded because f maps the trivial valuation of $k(X')$ to v_{triv}, and f^{an} is surjective because a valuation of the residue field at a point of X extends to the residue field at any point of X' above it.

The maximum principle in (6) is [BJ22, Proposition 4.12], and the vanishing at v_{triv} of a homogeneous function is immediate, v_{triv} being a fixed point of the scaling action. When

applying nef decreasing stability, we exclude the identically $-\infty$ limit, usually by evaluating at v_{triv}.

Remark 2.2. The compactness lemma [BJ22, Lemma 4.6] used to prove [BJ22, Theorem 4.5] is false when the approximating functions may take the value $-\infty$. In its proof, the strict inequality (4.5) fails at a common $-\infty$ point; the exponential difference used there then has limit zero, rather than a strictly negative limit.

Here is a counterexample to the abstract lemma. Let $K = [0,1]$, $S = \mathbf{Z}_{>0}$, $T = \mathbf{R}$, $\rho(n) = 1/n$, and

$$\mathcal{F}_n = \{x \longmapsto \log|x - 1/n| + c : c \in \mathbf{R}\}, \qquad \log 0 = -\infty.$$

These are families of continuous extended-real-valued functions, stable under constants. In the notation of that lemma, the constant function $-\infty$ belongs to $\widetilde{\mathcal{F}}_{1/n}$ for every n, by letting $c \downarrow -\infty$; hence it belongs to the iterated decreasing closure at 0. On the other hand, a decreasing net drawn from the $\mathcal{F}_n$ can involve only one index n: a common successor of terms with different indices would have to have two distinct poles, whereas every member of every $\mathcal{F}_n$ has exactly one. Thus no such net has $\rho(n) \to 0$, and $\widetilde{\mathcal{F}}_0 = \varnothing$.

This counterexample concerns the abstract diagonalization lemma, not the statement of [BJ22, Theorem 4.5] for psh functions. It does, however, leave that proof of decreasing stability incomplete in classes where singular Fubini–Study approximants are needed. Any use of that stability must therefore be supplied with an independent argument; the counterexample itself neither proves nor disproves it.

The arguments preceding the main theorem use decreasing stability only in the nef and homogeneous cases. The unibranch case is deduced afterwards in Corollary 8.4. The remaining limits used below are handled by the defining Fubini–Study approximation or by Lemma 7.4; normalization descent is proved first for homogeneous functions and then by reconstruction from their slices. We do not assert the unrestricted version of [BJ22, Theorem 4.5] on non-unibranch varieties in non-nef classes.

Lemma 2.3. *Let X be an integral projective variety. If $\theta_i \to \theta$ and a net $u_i \in \mathrm{PSH}_{\mathrm{hom}}(X, \theta_i)$ decreases pointwise to u, then $u \in \mathrm{PSH}_{\mathrm{hom}}(X, \theta)$.*

Proof. We first record an eventual domination property. Suppose $p_j \in \mathcal{H}_{\mathrm{hom}}(X, L_j)$ decrease to p, and q is a homogeneous Fubini–Study function with $p \le q$. For each rational $0 < a < 1$, we claim that $p_j \le aq$ eventually. On the compact set $\{q = -1\}$, the increasing open sets $\{p_j < -a\}$ cover; hence one index works throughout this set. Homogeneity gives the inequality wherever $-\infty < q < 0$, and it is automatic where $q = 0$. Write Z for the common zero locus of sections defining q. At the trivial semivaluation of the generic point of each irreducible component of Z, the homogeneous Fubini–Study values p_j are either 0 or $-\infty$ and decrease to $-\infty$. Thus they are eventually $-\infty$. Their defining sections then vanish on that component, so $p_j = -\infty$ on its whole analytification. There are finitely many components, which proves the claim also on $\{q = -\infty\}$.

Choose homogeneous Fubini–Study regularizations $p_{i,j} \downarrow u_i$, with classes $\eta_{i,j} \to \theta_i$, as justified above. Fix a norm on $N^1(X)_{\mathbf{R}}$. For triples $\alpha = (i, j, b)$ with rational $0 < b < 1$, put

$$F_\alpha = bp_{i,j}, \qquad e_\alpha = \|\eta_{i,j} - \theta_i\| + 1 - b.$$

Order the triples by increasing i, decreasing F_α and decreasing e_α. This preorder is directed. Indeed, given two triples, choose i_3 after their outer indices and rational $b_3 > \max\{b_1, b_2\}$ sufficiently close to 1. Since $u_{i_3} \le p_{i_k,j_k}$, the preceding domination property gives, for all sufficiently large j,

$$p_{i_3,j} \le (b_k/b_3)p_{i_k,j_k} \qquad (k = 1, 2).$$

Increasing j also makes $\|\eta_{i_3,j} - \theta_{i_3}\|$ as small as required. Thus a common successor exists with arbitrarily small error.

The resulting Fubini–Study net decreases, and $F_\alpha \geq u$, since $p_{i,j} \leq 0$. Its infimum is u: at any given point, first choose i, then j, and then b close to 1; at a $-\infty$ value use arbitrary finite thresholds. The errors tend to zero and the outer indices are cofinal, so the classes $b\eta_{i,j}$ tend to θ. Definition 4.1 of [BJ22] proves the assertion. Homogeneity passes to the pointwise limit, and $u(v_{\mathrm{triv}}) = 0$ excludes the identically $-\infty$ case. □

Recall from the introduction that $\mathrm{EP}_X(\theta)$ denotes the envelope property of θ. It has an increasing-net formulation which is convenient for descent arguments.

Proposition 2.4. *The envelope property $\mathrm{EP}_X(\theta)$ is equivalent to the following assertion: for every bounded-above increasing net (φ_i) in $\mathrm{PSH}(X,\theta)$ there is a (necessarily unique) $\varphi \in \mathrm{PSH}(X,\theta)$ with*

$$\varphi = \sup_i \varphi_i \quad \text{pointwise on } X^{\mathrm{div}}.$$

In this case $\varphi = \sup^*_i \varphi_i$; in particular φ is homogeneous if all φ_i are.

If the envelope property holds, [BJ22, Theorem 5.6] gives $(\sup_i \varphi_i)^* = \sup_i \varphi_i$ on X^{div}, which proves the asserted implication. Conversely, it suffices to consider the increasing net of finite maxima of an arbitrary bounded-above family. By Proposition 2.1(2), $\varphi \geq \varphi_i$ everywhere, so $\varphi \geq \sup^*_i \varphi_i$, and $\varphi \leq \sup^*_i \varphi_i$ since this holds on X^{div}. The scaling action is by homeomorphisms of X^{an}, so it commutes with pointwise suprema and usc regularization; this gives the last assertion.

2.3 Envelopes in ample classes

For an ample $L \in \mathrm{Pic}(X)_{\mathbf{Q}}$ and a bounded function $\phi\colon X^{\mathrm{an}} \to \mathbf{R}$, the *psh envelope* is the pointwise supremum

$$\mathrm{P}_L(\phi) \coloneqq \sup\{\tau \in \mathrm{PSH}(X,L) : \tau \leq \phi\},$$

without usc regularization. It is increasing in ϕ, and $\mathrm{P}_L(\phi + c) = \mathrm{P}_L(\phi) + c$ for $c \in \mathbf{R}$.

Rational piecewise linear (PL) functions on X^{an} are described by test configurations [BJ22, §2]. For the model description in this paragraph, assume that X is normal. Let L be a line bundle and $(\mathcal{X},\mathcal{L})$ a test configuration for (X,L) with $\mathcal{L}$ a line bundle. Passing to a higher test configuration, we may assume that $\mathcal{X}$ is normal and dominates $S = X \times \mathbf{A}^1$ through a projective birational morphism $\mu\colon \mathcal{X} \to S$ that is an isomorphism outside the central fibre. Then $\mathcal{L} = \mu^* L + D$ for a Cartier divisor D supported on the central fibre, and the associated PL function is $\phi_{\mathcal{L}} = \phi_D$. By [BJ22, Theorem 2.31], after multiplying the class and the function by a common positive integer, every rational PL function admits such a description. Replacing $\mathcal{L}$ by $\mathcal{L} + c\,\mathcal{X}_0$, with $\mathcal{X}_0 = \mathrm{div}(t)$ the central fibre and $c \in \mathbf{Z}$, adds c to $\phi_{\mathcal{L}}$.

Proposition 2.5. *Let $L \in \mathrm{Pic}(X)_{\mathbf{Q}}$ be ample.*

(1) $\mathrm{EP}_X(c_1(L))$ holds if and only if $\mathrm{P}_L(\phi)$ is L-psh for every rational PL function ϕ on X^{an}.

(2) Let L be a line bundle and $(\mathcal{X},\mathcal{L})$ a test configuration for (X,L) with $\mathcal{L}$ a line bundle, and let $\mathfrak{a}_m \subseteq \mathcal{O}_{\mathcal{X}}$ be the base ideal of $m\mathcal{L}$, generated by $H^0(\mathcal{X}, m\mathcal{L})$. Then

$$\mathrm{P}_L(\phi_{\mathcal{L}}) = \phi_{\mathcal{L}} + \sup_{m\geq 1} \frac{1}{m}\phi_{\mathfrak{a}_m} = \phi_{\mathcal{L}} + \lim_{m\to\infty} \frac{1}{m}\phi_{\mathfrak{a}_m} \qquad \text{pointwise on } X^{\mathrm{an}}.$$

Part (2) is [BJ22, Lemma 5.22] (see also [BJ24, Lemmas 1.5 and 1.6]). For (1), we give the PL-obstacle reduction in [BJ24, Lemma 1.3] explicitly. If the envelope property holds,

then the usc regularization of $\mathrm{P}_L(g)$ is L-psh and remains bounded above by the continuous obstacle g; hence it equals $\mathrm{P}_L(g)$. Conversely, assume that $\mathrm{P}_L(g)$ is L-psh for every rational PL function g. Let f be the supremum of a nonempty bounded-above family of L-psh functions, and let $\mathcal{G}$ be the rational PL majorants of f. This family is nonempty and stable under finite minima. Since rational PL functions are uniformly dense in $C^0(X^{\mathrm{an}})$ [BJ22, Theorem 2.2], and continuous majorants recover the usc regularization on the compact Hausdorff space X^{an}, we have $\inf_{g\in\mathcal{G}} g = f^*$. Ordered by decreasing obstacles, $\mathrm{P}_L(g)$ is a decreasing net of L-psh functions, bounded below by every member of the original family. PSH2 in the ample class $c_1(L)$ gives a psh limit u. Since $f \le u \le f^*$ and u is usc, $u = f^*$. This proves (1).

2.4 Fubini–Study classes under finite morphisms

Let $h\colon T \to X$ be a finite surjective morphism of integral projective varieties. The following lemma represents Fubini–Study functions on T in line bundles pulled back from X.

Lemma 2.6. *Let $h\colon T \to X$ be a finite surjective morphism of integral projective varieties and let $\theta \in N^1(X)_{\mathbf{R}}$. Suppose*

$$\varphi_i \in \mathcal{H}^{\mathrm{gf}}_{\mathbf{Q}}(T, M_i), \qquad c_1(M_i) \longrightarrow h^*\theta$$

is a sequence. After discarding finitely many initial terms, there are $L_i \in \mathrm{Pic}(X)_{\mathbf{Q}}$ with $c_1(L_i) \to \theta$ such that the same functions belong to $\mathcal{H}^{\mathrm{gf}}_{\mathbf{Q}}(T, h^*L_i)$, presented by sections of $h^*(mL_i)$ with mL_i an honest line bundle on X.

Proof. Fix an ample line bundle A on X. Since h^*A is ample, choose positive rational $\epsilon_i \to 0$ sufficiently slowly that $c_1(M_i) - h^*\theta = o(\epsilon_i)$. Choose rational line bundles L_i with $c_1(L_i) = \theta + \epsilon_i c_1(A) + o(\epsilon_i)$. The space $N^1(T)_{\mathbf{R}}$ is finite dimensional, and the ample cone is open by Kleiman's criterion. Putting $P_i \coloneqq h^*L_i - M_i$, we have $\epsilon_i^{-1} c_1(P_i) \to h^*c_1(A)$, so P_i is ample for all large i.

Fix i and take r sufficiently divisible that rP_i is an honest globally generated line bundle, with generators (t_a), and that rL_i is an honest line bundle on X. At every semivaluation, all their local coefficients are regular at the center and some coefficient is a unit. Thus $r^{-1}\max_a \log|t_a| = 0$ on T^{an}. Writing $\varphi_i = m^{-1}\max_j\{\log|s_j| + \lambda_j\}$ gives

$$\varphi_i = \frac{1}{rm}\max_{j,a}\{\log|s_j^{\otimes r}\otimes t_a^{\otimes m}| + r\lambda_j\}, \qquad s_j^{\otimes r}\otimes t_a^{\otimes m} \in H^0(T, rmh^*L_i).$$

This equality also holds at common zeros of the s_j, since the t_a have no common zero, and it proves the claim. □

3 The smooth case

In this section X is a smooth projective variety over k and $n = \dim X$.

Theorem 3.1. *Let X be a smooth projective variety over k. Then every nef class $\theta \in N^1(X)_{\mathbf{R}}$ has the envelope property.*

For $\mathrm{char}\, k = 0$ or $n \le 2$, this is [BJ22, Theorem 5.20]. We therefore assume $\mathrm{char}\, k = p > 0$. The proof occupies the rest of this section. Starting from the base-ideal formula for a PL envelope, we construct a decreasing approximation using asymptotic test ideals on the smooth scheme $S \coloneqq X \times \mathbf{A}^1$. A relative version of Mustaţă's generation theorem controls the classes of the approximants (Lemma 3.2), and a Frobenius-trace bound identifies their divisorial limit (Lemma 3.3). For the approach using test ideals on a resolution of the test configuration, see [BJ24, §2.1] and [GJKM19].

Reduction

Choose rational ample classes $\theta_j \to \theta$ with $\theta_j - \theta$ ample, which is possible since the ample cone is open and rational classes are dense. By [BJ22, Lemma 5.9], it suffices to prove $\mathrm{EP}_X(\theta_j)$ for all j. We may therefore assume $\theta = c_1(L)$ with $L \in \mathrm{Pic}(X)_{\mathbf{Q}}$ ample, and by Proposition 2.5(1) it suffices to show that $\mathrm{P}_L(\phi)$ is L-psh for every rational PL function ϕ. Since $\mathrm{P}_{mL}(m\phi) = m\mathrm{P}_L(\phi)$ and $\mathrm{P}_L(\phi + c) = \mathrm{P}_L(\phi) + c$, the discussion before Proposition 2.5 allows us to assume that L is a very ample line bundle and that $\phi = \phi_D$ for a test configuration

$$\mu\colon \mathcal{Y} \longrightarrow S, \qquad \mathcal{L} = \mu^* L + D, \qquad D \le 0,$$

where $\mathcal{Y}$ is normal, μ is projective, birational and an isomorphism outside the central fibre, and D is a Cartier divisor supported on the central fibre. Line bundles on X and their pullbacks to S are denoted by the same symbols. Let $\mathfrak{a}_m \subseteq \mathcal{O}_{\mathcal{Y}}$ be the base ideal of $m\mathcal{L}$. By Proposition 2.5(2),

$$\mathrm{Q} \coloneqq \mathrm{P}_L(\phi_D) = \phi_D + \sup_{m \ge 1} \frac{1}{m}\phi_{\mathfrak{a}_m} = \phi_D + \lim_{m\to\infty} \frac{1}{m}\phi_{\mathfrak{a}_m}. \tag{3.1}$$

In particular $\mathrm{Q} \le \phi_D$. We shall construct a decreasing sequence of psh functions whose limit agrees with Q on X^{div}.

The graded ideals J_m

Since S is normal and μ is proper and birational, $\mu_*\mathcal{O}_{\mathcal{Y}} = \mathcal{O}_S$. As $D \le 0$, this gives $H^0(\mathcal{Y}, m\mathcal{L}) \subseteq H^0(\mathcal{Y}, \mu^*(mL)) = H^0(S, mL)$. Let $J_m \subseteq \mathcal{O}_S$ be the ideal such that $J_m \otimes \mathcal{O}_S(mL)$ is the subsheaf of $\mathcal{O}_S(mL)$ generated by $H^0(\mathcal{Y}, m\mathcal{L})$. Then

$$J_m J_r \subseteq J_{m+r},$$

every power $J_q^r \otimes \mathcal{O}_S(qrL)$ is globally generated, and $J_m \neq 0$: for $0 \neq s \in H^0(X, mL)$ and N large, $t^N s \in H^0(\mathcal{Y}, m\mathcal{L})$, since $-mD$ is supported on the central fibre, where t vanishes. Both $(J_m\mathcal{O}_{\mathcal{Y}}) \otimes \mu^*\mathcal{O}_S(mL)$ and $\mathfrak{a}_m \otimes \mathcal{O}_{\mathcal{Y}}(m\mathcal{L})$ are the subsheaf of $\mu^*\mathcal{O}_S(mL)$ generated by $H^0(\mathcal{Y}, m\mathcal{L})$. Hence

$$J_m \cdot \mathcal{O}_{\mathcal{Y}} = \mathfrak{a}_m \cdot \mathcal{O}_{\mathcal{Y}}(mD)$$

as ideals of $\mathcal{O}_{\mathcal{Y}}$, where $\mathcal{O}_{\mathcal{Y}}(mD) \subseteq \mathcal{O}_{\mathcal{Y}}$ is the ideal of the effective divisor $-mD$. Evaluating $\sigma(v)$ on both sides gives $\frac{1}{m}\phi_{J_m} = \phi_D + \frac{1}{m}\phi_{\mathfrak{a}_m}$ on X^{an}, and (3.1) becomes

$$\mathrm{Q} = \sup_{m \ge 1} \frac{1}{m}\phi_{J_m} = \lim_{m\to\infty} \frac{1}{m}\phi_{J_m}. \tag{3.2}$$

Asymptotic test ideals

The smooth scheme S is F-finite since k is perfect. For a nonzero coherent ideal $I \subseteq \mathcal{O}_S$ and $c \in \mathbf{Q}_{\ge 0}$, its test ideal $\tau(I^c)$ is given by the stable Frobenius-trace image

$$\mathrm{Tr}^e\Big(F_*^e\big(I^{\lceil cp^e \rceil} \otimes \omega_S\big)\Big) = \tau(I^c) \otimes \omega_S \qquad (e \gg 0), \tag{3.3}$$

where F is Frobenius and $\mathrm{Tr}^e : F_*^e\omega_S \to \omega_S$ is its Grothendieck trace [Stacks, Tag 0FK6]; see [Mus13, §3] for this description and its stabilization.

For $m \ge 1$, the *asymptotic test ideal* $B_m \coloneqq \tau(J_\bullet^m)$ equals $\tau(J_q^{m/q})$ for every sufficiently divisible q, and

$$J_m \subseteq B_m, \qquad B_{m\ell} \subseteq B_m^\ell \qquad (\ell \ge 1), \tag{3.4}$$

the second inclusion being the subadditivity of asymptotic test ideals. See Hara–Yoshida [HY03] and Blickle–Mustaţă–Smith [BMS08] for the local theory, [ST12] for a survey, [Mus13, §3, Proposition 3.1] for the asymptotic construction in the form used here, and [GJKM19, §4] for the compatibility between them. Put

$$u_m \coloneqq \frac{1}{m}\phi_{B_m} \qquad (m \geq 1).$$

Since $I \subseteq J$ implies $\phi_I \leq \phi_J$, (3.4) gives $\frac{1}{m\ell}\phi_{J_{m\ell}} \leq u_{m\ell} \leq u_m$ for $\ell \geq 1$. Letting $\ell \to \infty$ and using (3.2) and $B_m \subseteq \mathcal{O}_S$, we obtain

$$\mathrm{Q} \leq u_m \leq 0 \qquad \text{on } X^{\mathrm{an}}. \tag{3.5}$$

A uniform twist

Fix a very ample line bundle H on X and put

$$A \coloneqq K_X + (n+1)H.$$

The following relative version of Mustaţă's uniform generation theorem [Mus13, Theorem A] supplies a twist independent of m.

Lemma 3.2. *For every $m \geq 1$, the sheaf $B_m(mL+A)$ is globally generated relative to $\mathbf{A}^1$, and*

$$u_m \in \mathcal{H}^{\mathrm{gf}}_{\mathbf{Q}}\left(X, L + \frac{1}{m}A\right). \tag{3.6}$$

Proof. Fix m and choose q sufficiently divisible that $B_m = \tau(J_q^{m/q})$. Put $r_e = \lceil mp^e/q \rceil$. For $e \gg 0$, (3.3) gives a surjection

$$F_*^e(J_q^{r_e} \otimes \omega_S) \twoheadrightarrow B_m \otimes \omega_S.$$

Since $J_q^{r_e}(qr_eL)$ is globally generated, it is a quotient of a finite sum of copies of $\mathcal{O}_S$. Apply the exact functor F_*^e to the corresponding surjection onto $J_q^{r_e} \otimes \omega_S$, compose with trace, and tensor by $mL + (n+1)H$. The projection formula gives a surjection

$$\bigoplus F_*^e\big(\mathcal{O}_S(K_S + (mp^e - qr_e)L + p^e(n+1)H)\big) \twoheadrightarrow B_m(mL+A). \tag{3.7}$$

Here $K_S = \mathrm{pr}_X^* K_X$ since $K_{\mathbf{A}^1}$ is trivial, so the displayed twists are exact line-bundle identities.

For $1 \leq i \leq n$, the higher direct image of a summand in (3.7) twisted by $-iH$ is

$$F^e_{\mathbf{A}^1,*}\Big(H^i\big(X, K_X + a_eL + p^e(n+1-i)H\big) \otimes_k k[t]\Big), \qquad a_e \coloneqq mp^e - qr_e \in \{1-q, \dots, 0\}.$$

This uses $\rho \circ F_S^e = F^e_{\mathbf{A}^1} \circ \rho$ for the projection $\rho\colon S \to \mathbf{A}^1$ and the cohomology formula for the product projection. Since $n+1-i \geq 1$, Serre vanishing [Har77, Chapter III, Theorem 5.2] for the finitely many residual bundles $K_X + a_eL$ kills these groups for all sufficiently large e; for $i > n$ they vanish by dimension. Relative Castelnuovo–Mumford regularity [Laz04, Example 1.8.24] shows that the source, and hence its quotient, is relatively globally generated. The necessary e may depend on m and q, but A is fixed.

Over the affine base, choose finitely many global generators $g_j = \sum_\lambda s_{j\lambda}t^\lambda$ of $B_m(mL+A)$, where $s_{j\lambda} \in H^0(X, mL+A)$ and $\lambda \in \mathbf{N}$. The Gauss-extension formula yields

$$u_m = \frac{1}{m}\max_{j,\lambda}\{\log|s_{j\lambda}| - \lambda\},$$

which is (3.6), also at common zeros. □

A divisorial bound for test ideals

Since S is smooth, the *log discrepancy* of a nontrivial divisorial valuation $w = r\,\mathrm{ord}_E$ on S, with E a prime divisor on a normal birational model $g\colon V \to S$ and $r \in \mathbf{Q}_{>0}$, is

$$A_S(w) \coloneqq r(1 + \mathrm{ord}_E(K_{V/S})) \in \mathbf{Q}_{>0};$$

it is independent of the model and homogeneous of degree one in w. We only need a valuative bound at divisorial valuations. It is a special case of the discrepancy bounds in [Can20], but it also admits the following direct proof using the Frobenius trace on top differential forms; only smoothness of S over the perfect field k is used.

Lemma 3.3. *Let w be a nontrivial divisorial valuation with center on S, let $I \subseteq \mathcal{O}_S$ be a nonzero ideal and let $c \in \mathbf{Q}_{\geq 0}$. Then*

$$w(\tau(I^c)) \geq c\,w(I) - A_S(w). \tag{3.8}$$

Proof. Fix a trivialization η of ω_S near the center and write $\mathrm{Tr}^e(f\eta) = \Phi_e(f)\eta$. We first prove that, for every $e \geq 1$ and every local function f,

$$w(\Phi_e(f)) \geq \frac{w(f)}{p^e} - \left(1 - \frac{1}{p^e}\right) A_S(w). \tag{3.9}$$

First take $w = \mathrm{ord}_E$ and choose a normal birational model $g\colon V \to S$ carrying E. Then $R = \mathcal{O}_{V,E}$ is a DVR, so V is regular at the generic point of E and hence, the regular locus being open and k perfect, smooth over k near it; the whole computation takes place in R. If η generates ω_S near the center, write

$$g^*\eta = h\xi, \qquad a \coloneqq \mathrm{ord}_E(h) = A_S(\mathrm{ord}_E) - 1 \geq 0,$$

with ξ a generator of top differential forms at R; the inequality holds because $g^*\eta$ is a regular top form there. For a rational top form α we write $\mathrm{ord}_E(\alpha)$ for $\mathrm{ord}_E(\alpha/\xi)$, so that $\mathrm{ord}_E(fg^*\eta) = \mathrm{ord}_E(f) + a$. On rational top forms Frobenius trace is the iterated Cartier operator: being compatible with localization, the traces of S and of V both restrict at the common generic point to the e-th Cartier operator of $\Omega^{\dim S}_{K/k}$, $K = k(S) = k(V)$, which depends on K alone; and $\mathrm{Tr}^e\colon F^e_*\omega_R \to \omega_R$ exists because R is regular and F-finite, so regular top forms are preserved. See [Mus13, §3] and [ST12] for the trace–Cartier description and its local coordinate formula.

Let z be a uniformizer and put $\ell = \lfloor (\mathrm{ord}_E(f) + a)/p^e \rfloor$. Then $fg^*\eta = z^{p^e\ell}\omega$ for a regular top form ω at R. Semilinearity and regularity of trace give

$$\mathrm{Tr}^e(fg^*\eta) = z^\ell\,\mathrm{Tr}^e(\omega), \qquad \mathrm{ord}_E\,\mathrm{Tr}^e(fg^*\eta) \geq \ell.$$

Since $\mathrm{Tr}^e(fg^*\eta) = \Phi_e(f)\,g^*\eta$ has ord_E equal to $\mathrm{ord}_E\,\Phi_e(f) + a$, this yields

$$\mathrm{ord}_E\,\Phi_e(f) \geq \left\lfloor \frac{\mathrm{ord}_E(f) + a}{p^e} \right\rfloor - a \geq \frac{\mathrm{ord}_E(f) - (p^e - 1)(a+1)}{p^e},$$

which is (3.9) for $w = \mathrm{ord}_E$ because $(p^e - 1)(a + 1) = (p^e - 1)A_S(\mathrm{ord}_E)$. Both sides are homogeneous of degree one in w, which extends the bound to every nontrivial divisorial w.

Finally, fix $e \gg 0$ and put $N = \lceil cp^e \rceil$. By (3.3), $\tau(I^c) = \{\Phi_e(f) : f \in I^N\}$ near the center of w. Every $f \in I^N$ satisfies $w(f) \geq N\,w(I) \geq cp^e\,w(I)$, since $w(I) \geq 0$, so (3.9) and $A_S(w) > 0$ give $w(\Phi_e(f)) \geq c\,w(I) - A_S(w)$. This proves (3.8). □

Conclusion of the proof

Take $m_j = j!$. By (3.4) the sequence (u_{m_j}) decreases, and by (3.5) it is bounded below by Q, which is finite at v_{triv} because $\mathrm{Q} \geq \frac{1}{m}\phi_{J_m}$ and $J_m \neq 0$. Thus Lemma 3.2 and the defining decreasing Fubini–Study approximation [BJ22, Definition 4.1] give

$$u \coloneqq \lim_j u_{m_j} \in \mathrm{PSH}(X, L).$$

Let $v \in X^{\mathrm{div}}$. Then $w \coloneqq \sigma(v)$ is a divisorial valuation on S (Section 2.1), nontrivial since $w(t) = 1$. Choose q sufficiently divisible that $B_m = \tau(J_q^{m/q})$. Since $\mathrm{Q} \geq \frac{1}{q}\phi_{J_q}$, that is, $w(J_q) \geq -q\,\mathrm{Q}(v)$, Lemma 3.3 gives $w(B_m) \geq -m\,\mathrm{Q}(v) - A_S(w)$. Together with (3.5), this gives

$$\mathrm{Q}(v) \leq u_m(v) \leq \mathrm{Q}(v) + \frac{A_S(w)}{m}.$$

Hence $u = \mathrm{Q}$ on X^{div}. Since $\mathrm{Q} \leq \phi_D$ and ϕ_D is continuous, divisorial comparison gives $u \leq \phi_D$ on X^{an}, so $u \leq \mathrm{P}_L(\phi_D) = \mathrm{Q} \leq u$ by (3.5). Thus $\mathrm{P}_L(\phi_D) = u$ is L-psh, which completes the proof of Theorem 3.1. □

4 Finite descent

Let $h\colon T \to X$ be a finite surjective morphism of integral projective varieties, with X normal. Field norms of sections of pulled-back line bundles push Fubini–Study functions on T forward to X; normality of X makes the norm coefficients regular, while the source T need not be normal. Divisorial comparison supplies the monotonicity needed for decreasing limits. This gives finite descent of the envelope property (Theorem 4.2).

Write $E = k(T)$, $K = k(X)$ and $d = [E : K]$. For $v \in X^{\mathrm{div}}$, including v_{triv}, the support of v is generic, so $K \hookrightarrow \mathcal{H}(v)$. Decompose the finite algebra

$$E \otimes_K \mathcal{H}(v) = \prod_{w|v} B_w, \qquad d(w/v) \coloneqq \dim_{\mathcal{H}(v)} B_w, \qquad \sum_{w|v} d(w/v) = d.$$

Here each B_w is local Artinian; its residue field, finite over the complete field $\mathcal{H}(v)$, has a unique extending absolute value, and $E \hookrightarrow B_w$ followed by reduction is injective. These give exactly the extensions w of v to E, with no separability assumption [AC, Ch. VI, §8].

Every such extension is a point of T^{div}. Indeed, if $v = c\,\mathrm{ord}_D$ on a normal projective model $X' \to X$, normalize X' in E. This is a finite normal model over X' and maps birationally to T; the extensions of v are positive rational multiples of the orders along the prime divisors over D. Over v_{triv} the only extension is the trivial valuation of E.

Proposition 4.1. *Let $L \in \mathrm{Pic}(X)_{\mathbf{Q}}$ and let*

$$\varphi = \frac{1}{m}\max_j\{\log|s_j| + \lambda_j\} \in \mathcal{H}^{\mathrm{gf}}_{\mathbf{Q}}(T, h^*L), \qquad s_j \in H^0(T, h^*(mL)).$$

The universal field norm of this system has coefficients $c_I \in H^0(X, mdL)$, $|I| = d$, not all zero. They define

$$\mathrm{Av}_h(\varphi) \coloneqq \frac{1}{md}\max_{|I|=d}\left\{\log|c_I| + \sum_j I_j\lambda_j\right\} \in \mathcal{H}^{\mathrm{gf}}_{\mathbf{Q}}(X, L). \tag{4.1}$$

At every divisorial point one has

$$\mathrm{Av}_h(\varphi)(v) = \frac{1}{d}\sum_{w|v} d(w/v)\varphi(w). \tag{4.2}$$

The construction is independent of the presentation and preserves pointwise order, including between functions presented in different line bundles.

Proof. On an affine $U = \operatorname{Spec} A \subset X$ trivializing mL with frame e, write $s_j = f_j h^* e$ with $f_j \in B \coloneqq \mathcal{O}_T(h^{-1}U)$. Introduce independent variables $\mathbf{T} = (T_j)$ and form

$$P_U(\mathbf{T}) = \mathrm{N}_{E(\mathbf{T})/K(\mathbf{T})}\left(\sum_j T_j f_j\right) = \sum_{|I|=d} c_{I,U}\mathbf{T}^I.$$

This is the determinant of multiplication on the free $K[\mathbf{T}]$-module $E[\mathbf{T}]$, so it is homogeneous of degree d. The linear form is integral over $A[\mathbf{T}]$, and its norm is integral over that ring and lies in $K[\mathbf{T}]$. Since $A[\mathbf{T}]$ is normal, $P_U \in A[\mathbf{T}]$. Replacing e by ge multiplies P_U by g^{-d}; hence its coefficients glue to sections of mdL. At the generic point the linear form is nonzero and has nonzero field norm, so some $c_I \neq 0$. Thus (4.1) defines a Fubini–Study function in the asserted class.

Fix $v \in X^{\mathrm{div}}$, take U containing its center, and extend $\mathcal{H}(v)$ isometrically to a complete algebraically closed field C. The finite C-algebra $E \otimes_K C$ is a product of local Artin algebras with residue field C. On a factor of length a, multiplication by $\sum_j T_j f_j$ has determinant equal to the a-th power of its residue linear form: on the filtration by powers of the maximal ideal it acts by that residue on each successive quotient. All embeddings of a residue field of B_w into C are isometric, and the local factors of $E \otimes_K C$ lying over w have total C-dimension $d(w/v)$, determinants being compatible with the base change $K[\mathbf{T}] \to C[\mathbf{T}]$. The *weighted Gauss norm* of $\sum_I q_I \mathbf{T}^I \in C[\mathbf{T}]$ is $\max_I\{|q_I| \prod_j \mathrm{e}^{\lambda_j I_j}\}$, equivalently the supremum norm on the Berkovich polydisc of polyradius $(\mathrm{e}^{\lambda_j})_j$; it is multiplicative, and therefore yields

$$\max_{|I|=d}\left\{\log|c_I|(v) + \sum_j I_j\lambda_j\right\} = \sum_{w|v} d(w/v)\max_j\{\log|s_j|(w) + \lambda_j\}.$$

Dividing by md proves (4.2). Two presentations therefore give psh functions with the same divisorial values, so divisorial comparison shows that they give the same function on X^{an}.

If $\varphi \leq \psi$ on T^{an}, then $\mathrm{Av}_h(\varphi) \leq \mathrm{Av}_h(\psi)$ on X^{div} by (4.2), hence on all of X^{an} by divisorial comparison, the left-hand side being psh and the right-hand side usc; this holds for functions presented in different numerical classes, since Proposition 2.1(2) requires nothing more. □

Theorem 4.2. *Let $h\colon T \to X$ be a finite surjective morphism of integral projective varieties, with X normal, and $\theta \in N^1(X)_{\mathbf{R}}$. If $\mathrm{EP}_T(h^*\theta)$ holds, then so does $\mathrm{EP}_X(\theta)$.*

Proof. Let (φ_i) be a bounded-above increasing net in $\mathrm{PSH}(X,\theta)$. By the envelope property upstairs and Proposition 2.4, there is $F \in \mathrm{PSH}(T, h^*\theta)$ with $F = \sup_i \varphi_i \circ h^{\mathrm{an}}$ on T^{div}. By Proposition 2.1(3) and Lemma 2.6, F is the limit of a decreasing sequence $F_j \in \mathcal{H}^{\mathrm{gf}}_{\mathbf{Q}}(T, h^*L_j)$ with $c_1(L_j) \to \theta$. By Proposition 4.1, the functions $\mathrm{Av}_h(F_j) \in \mathcal{H}^{\mathrm{gf}}_{\mathbf{Q}}(X, L_j)$ decrease on X^{an}. The only extension of v_{triv} is the trivial valuation of E, so $\mathrm{Av}_h(F_j)(v_{\mathrm{triv}}) = F_j(v_{\mathrm{triv}}) \to F(v_{\mathrm{triv}}) \in \mathbf{R}$, and [BJ22, Definition 4.1] gives $G \coloneqq \lim_j \mathrm{Av}_h(F_j) \in \mathrm{PSH}(X,\theta)$ directly, since these approximants are themselves Fubini–Study functions with classes tending to θ. For $v \in X^{\mathrm{div}}$, every extension w of v lies in T^{div} and satisfies $h(w) = v$, so passing to the limit in (4.2) gives

$$G(v) = \frac{1}{d}\sum_{w|v} d(w/v)F(w) = \frac{1}{d}\sum_{w|v} d(w/v)\sup_i \varphi_i(v) = \sup_i \varphi_i(v).$$

Proposition 2.4 proves $\mathrm{EP}_X(\theta)$. □

5 Local regularization on normal varieties

We prove that, over a countable field, a function on a normal projective variety is psh in an ample class whenever its pullback to a normal projective modification is psh (Corollary 5.26).

This is a non-Archimedean analogue of the theorem of Fornæss–Narasimhan [FN80]. Apart from this corollary, only Lemmas 5.2 and 5.4 are used later.

The argument follows Wu [Wu26], with the modifications needed in arbitrary characteristic. On an affine chart, we consider *regularizable* functions, defined as decreasing limits of affine Fubini–Study functions. Polynomial hulls and norms along generically étale finite projections extend regularizability from the complement of a closed subset (Theorem 5.25). For the global application, regularizability away from the exceptional locus follows directly from the psh pull-back. We cite the local extension and hull arguments of [Wu26], recording minor corrections in footnotes. We give proofs where the passage to arbitrary characteristic or the form needed here requires a further argument.

Conventions. Throughout this section k is countable, algebraically closed and trivially valued. Closed subsets carry their reduced induced structure. Countability enters only through the countability of $k[Y]$ for an affine variety Y, in Lemma 5.14(2) and Lemma 5.18; in particular Y^{an} is second countable. For a variety Y, the set $Y^{\mathrm{div}} \subseteq Y^{\mathrm{an}}$ of divisorial points consists of v_{triv} and of the valuations $c\,\mathrm{ord}_E$ of $k(Y)$ described in Section 2.1; thus $Y^{\mathrm{div}} = P^{\mathrm{div}}$ if Y is an open subset of a projective variety P. For a regular function f we put $\log|f|(y) = -y(f)$. If M is a line bundle on a projective variety P and $0 \neq s \in H^0(P, M)$, we write $P_s \subseteq P$ for the open subset where s does not vanish. Thus $P_s^{\mathrm{an}} = \{\log|s| > -\infty\}$, and $\log|\sigma| - \log|s| = \log|\sigma/s|$ on P_s^{an} for every $\sigma \in H^0(P, M)$.

5.1 Local psh functions

In this subsection F is a complete non-Archimedean field, which will be k or a complete extension of k, and V is a scheme of finite type over F. Complete extensions of k are used in Lemma 5.5 and Section 5.5.

Definition 5.1. Let $U \subseteq V^{\mathrm{an}}$ be open.

(1) A function $u\colon U \to \mathbf{R}$ is *convex PL* if every point of U has an open neighbourhood $W \subseteq U$ with the following property: there are a Zariski open subset $V_0 \subseteq V$ with $W \subseteq V_0^{\mathrm{an}}$, invertible functions $f_1, \dots, f_r \in \mathcal{O}(V_0)^\times$ and rational numbers a_{ij}, b_j such that

$$u = \max_{1 \le j \le N} \Big(\sum_{i=1}^{r} a_{ij} \log|f_i| + b_j \Big) \qquad \text{on } W.$$

(2) A function on U is *locally PL* if it is locally the difference of two convex PL functions.

(3) A function $u\colon U \to [-\infty, \infty)$ is *locally psh* if every point of U has an open neighbourhood $W \subseteq U$ on which u is the pointwise limit of a decreasing sequence of convex PL functions on W.

Convex PL functions are continuous, and locally psh functions are upper semicontinuous; the latter may be identically $-\infty$ on some open subsets. Definition 5.1 is a variant of [Wu26, Definitions 3.7 and 3.8], using invertible functions and explicit finite maxima.[1]

Lemma 5.2. *Let $U \subseteq V^{\mathrm{an}}$ be open.*

(1) Finite maxima, finite sums, positive rational multiples and rational translates of convex PL (resp. locally psh) functions on U are convex PL (resp. locally psh). If f is invertible on a Zariski open subset $V_0 \subseteq V$, then adding a rational multiple of $\log|f|$ preserves both classes on $U \cap V_0^{\mathrm{an}}$.

[1] In [Wu26, Definition 3.1(3)], the supporting affine function must touch the graph: replace the displayed condition by $u(x) = l(x)$ and $u \ge l$ near x. Our formulation also avoids conventions at infinite tropical coordinates.

(2) *Let $g\colon V'^{\mathrm{an}} \to V^{\mathrm{an}}$ be the analytification of a morphism $V' \to V$ of schemes of finite type over F, or the natural map $V'^{\mathrm{an}}_{F'} \to V^{\mathrm{an}}$ for a complete extension F'/F. If u is convex PL (resp. locally psh) on U, then so is $u \circ g$ on $g^{-1}(U)$.*

(3) *Let $U' \subseteq U$ be open, u locally psh on U and v locally psh on U', and assume that the closure in U of $\{x \in U' : v(x) > u(x)\}$ is contained in U'. Then the function equal to $\max\{u, v\}$ on U' and to u on $U \setminus U'$ is locally psh on U, and it is convex PL if u and v are.*

(4) *Every locally psh function is classically psh in the sense of [GR25, Definition 4.1]. In particular its pullback under a morphism from an analytic curve is subharmonic, and it satisfies the local maximum principle [GR25, Theorem 6.1].*

Proof. (1) A sum of two finite maxima is the maximum of the pairwise sums, and maxima and sums of decreasing sequences decrease to the maxima and sums of the limits. Moreover, if u_j decrease to u and rational q_j increase to $c \in \mathbf{R}$, then $u_j - q_j$ decrease to $u - c$; so locally psh functions are also stable under real translates. (2) Invertible functions pull back to invertible functions, and $\log|f|\circ g = \log|f\circ g|$. (3) On U' this is (1). A point of $U\setminus U'$ has a neighbourhood disjoint from the closure in question, on which the function equals u. (4) For f invertible and analytic, $\log|f|$ and $-\log|f| = \log|f^{-1}|$ are classically psh [GR25, Proposition 4.2(2)], and so are constants. Hence convex PL functions are classically psh by the cone property, and so are locally psh functions, by stability under decreasing limits and the sheaf property [GR25, Proposition 4.2(1), (3), (4)]. On curves, classically psh functions are the subharmonic ones [GR25, §4.1.2]. □

We now compare these notions with global psh functions. Let P be a projective variety over k and $L \in \mathrm{Pic}(P)_{\mathbf{Q}}$ semiample. We call (m, s) an *admissible pair* for L if $m \geq 1$, mL is a globally generated line bundle and $0 \neq s \in H^0(P, mL)$. Choose generators σ_j of mL with s among them. Then $\max_j \log|\sigma_j| = 0$, so

$$-\frac{1}{m}\log|s| = \frac{1}{m}\max_j \log|\sigma_j/s| \qquad \text{on } P_s^{\mathrm{an}}. \tag{5.1}$$

Near a point $x \in P_s^{\mathrm{an}}$ the terms with σ_j/s vanishing at the support of x may be omitted, and the remaining σ_j/s are invertible near that support. Hence $-m^{-1}\log|s|$ is convex PL on P_s^{an}. If (m', s') is another admissible pair, then $m^{-1}\log|s| - m'^{-1}\log|s'| = (mm')^{-1}\log|s^{m'}/s'^{m}|$, where $s^{m'}/s'^{m}$ is invertible on $P_s \cap P_{s'}$.

Definition 5.3. Let $U \subseteq P^{\mathrm{an}}$ be open. A function $\varphi\colon U \to [-\infty, \infty)$ is *locally L-psh* (resp. *locally L-convex*) if $\varphi - m^{-1}\log|s|$ is locally psh (resp. convex PL) on $U \cap P_s^{\mathrm{an}}$ for every admissible pair (m, s) for L.

By the last observation and Lemma 5.2(1), it suffices that each point of U lies in some P_s^{an} near which the condition holds for (m, s).

Lemma 5.4. *Let P be a projective variety and $L \in \mathrm{Pic}(P)_{\mathbf{Q}}$ semiample.*

(1) *Every $\varphi \in \mathcal{H}^{\mathrm{gf}}_{\mathbf{Q}}(P, L)$ defined by sections without common zero is locally L-convex.*

(2) *Every $\Phi \in \mathrm{PSH}(P, L)$ is locally L-psh.*

Proof. (1) Write $\varphi = N^{-1}\max_\ell\{\log|\tau_\ell| + \lambda_\ell\}$ with $\tau_\ell \in H^0(P, NL)$ without common zero. On P_s^{an} we have $\varphi - m^{-1}\log|s| = (Nm)^{-1}\max_\ell\{\log|\tau_\ell^m/s^N| + m\lambda_\ell\}$, a maximum of logarithms of regular functions on P_s plus constants. Near a point we may omit the terms that vanish at its support, and at least one term remains since the τ_ℓ have no common zero.

(2) Since L is nef, [BJ22, Corollary 12.18(ii)] gives $M_j \in \mathrm{Pic}(P)_{\mathbf{Q}}$ with $c_1(M_j) \to c_1(L)$ and $\Phi_j \in \mathcal{H}^{\mathrm{gf}}_{\mathbf{Q}}(P, M_j)$, defined by sections without common zero, decreasing to Φ. Fix a globally generated ample line bundle H. Passing to a subsequence, choose rational $\epsilon_j \downarrow 0$ such that $L+\epsilon_j H - M_j$ is ample. Tensoring the sections defining Φ_j with a generating family of a multiple of $L+\epsilon_j H - M_j$, whose homogeneous Fubini–Study function is zero, shows that $\Phi_j \in \mathcal{H}^{\mathrm{gf}}_{\mathbf{Q}}(P, L+\epsilon_j H)$, still defined by sections without common zero. Given $x \in P^{\mathrm{an}}_s$, choose $t \in H^0(P,H)$ not vanishing at the support of x and put $W := (P_s \cap P_t)^{\mathrm{an}}$. By (1), applied to $L+\epsilon_j H$ and the admissible pair given by a common power of s and t, the functions $\Phi_j - m^{-1}\log|s| - \epsilon_j \log|t|$ are convex PL on W. They decrease to $\Phi - m^{-1}\log|s|$, because $-\log|t| \geq 0$ and $\epsilon_j \downarrow 0$. □

We write $\mathrm{PL}(P^{\mathrm{an}})$ for the $\mathbf{Q}$-vector space of rational PL functions on P^{an} [BJ22, §2]; it is stable under finite maxima and minima, and it is dense in $C^0(P^{\mathrm{an}})$ [BJ22, Theorem 2.2].

The following is a special case of [Wu26, Theorem 3.23], which relies on the positivity criterion of Chambert-Loir and Ducros [CLD25]. Since that criterion is formulated for nontrivially valued fields, we explain the passage from the trivially valued setting.

Lemma 5.5. *Let P be a normal projective variety and $L \in \mathrm{Pic}(P)_{\mathbf{Q}}$ ample. Every locally L-convex function $\varphi\colon P^{\mathrm{an}} \to \mathbf{R}$ belongs to $\mathrm{PSH}(P,L)$.*

Proof. By (5.1), φ is locally PL. We first show that φ agrees near each point x with a function in $\mathrm{PL}(P^{\mathrm{an}})$. By the definition of locally PL functions, it suffices to treat $\log|f|$, where f is a rational function that is regular and invertible at the support ξ of x. Fix an ample line bundle A and let $\mathfrak{d}$ be the ideal of denominators of f; it is trivial at ξ. For m large choose $s_0 \in H^0(P, mA\otimes\mathfrak{d})$ with $s_0(\xi) \neq 0$; then $s_1 := fs_0 \in H^0(P, mA)$ and $s_1(\xi) \neq 0$. Complete s_0, s_1 to a generating family (s_j) of mA. Choose one rational $\lambda > \max_{i=0,1}\{-\log|s_i|(x)\}$. For $i = 0, 1$, the Fubini–Study function $\psi_i := m^{-1}\max\{\log|s_i| + \lambda, \max_{j\neq i}\log|s_j|\}$ equals $m^{-1}(\log|s_i| + \lambda)$ near x, since $\log|s_j| \leq 0$ for all j. Hence $\log|f| = m(\psi_1 - \psi_0)$ near x.

By compactness, P^{an} is covered by the interiors of finitely many compact sets Q_i such that $\varphi = G_i$ on an open neighbourhood V_i of Q_i, with $G_i \in \mathrm{PL}(P^{\mathrm{an}})$. Choose continuous $\chi_i\colon P^{\mathrm{an}} \to [0,1]$ equal to 1 on Q_i with support in V_i, and $\tau_i \in \mathrm{PL}(P^{\mathrm{an}})$ with $|\tau_i - \chi_i| \leq 1/8$. Then $b_i := \max\{0, \min\{1, 2\tau_i - 1/2\}\} \in \mathrm{PL}(P^{\mathrm{an}})$ takes values in $[0,1]$, equals 1 on Q_i, and vanishes outside $\{\chi_i \geq 1/8\} \subseteq V_i$. Let $M \in \mathbf{Q}$ bound all $|G_i|$ and $|\varphi|$, which are continuous, and put $G := \min_i\{G_i + 4M(1-b_i)\} \in \mathrm{PL}(P^{\mathrm{an}})$. Let $y \in \operatorname{int} Q_{i_0}$. The term of index i_0 equals $\varphi(y)$. If $b_i(y) > 0$, then $y \in V_i$ and the term of index i is at least $\varphi(y)$; if $b_i(y) = 0$, it is at least $-M + 4M \geq \varphi(y)$. Thus $\varphi = G \in \mathrm{PL}(P^{\mathrm{an}})$.

Replacing L and φ by a common positive multiple, we may write $\varphi = \phi_{\mathcal{L}}$ for a line bundle $\mathcal{L}$ on a normal test configuration $\mathcal{P}$ of (P,L) dominating $P \times \mathbf{A}^1$ [BJ22, Theorem 2.31]. By [BJ22, Definition 3.2 and Lemma 4.3] it suffices to show that $\mathcal{L}$ has nonnegative degree on every integral curve C contained in the central fibre $\mathcal{P}_0$.

Let $K = k((\varpi))$ with $|\varpi| = \mathrm{e}^{-1}$, let $\mathfrak{P}$ be the formal completion of $\mathcal{P} \times_{\mathbf{A}^1} \operatorname{Spec} k[[\varpi]]$ along its special fibre $\mathcal{P}_0$, and let $\mathfrak{L}$ be the induced formal line bundle. This is a formal model of (P_K, L_K) and defines an integral PL metric on L_K. Let $r\colon P^{\mathrm{an}}_K \to P^{\mathrm{an}}$ be the restriction map. Then $\varphi \circ r$ is the model function of $\mathfrak{L}$ relative to the trivial model of L. Indeed, both sides are additive in $\mathcal{L}$, so it suffices to treat $\log|\mathfrak{a}|$ for a $\mathbf{G}_m$-invariant ideal $\mathfrak{a} = \sum_\lambda \mathfrak{a}_\lambda t^\lambda$ on $P \times \mathbf{A}^1$, which is generated by elements $a\,t^\lambda$ with $a \in \mathfrak{a}_\lambda$; at $w \in P^{\mathrm{an}}_K$ both sides then equal $\max_\lambda\{\log|\mathfrak{a}_\lambda|(r(w)) - \lambda\}$. By hypothesis and Lemma 5.2(2), near every point of P^{an}_K the metric has a local potential of the form $h \circ (\log|f_1|, \ldots, \log|f_r|)$ with f_i invertible and h convex. Such a function is psh in the sense of [CLD25] by [CLD25, Corollaire 18.8.9], so the curvature current of the metric is positive [CLD25, Proposition 8.3.1]. By [CLD25, Théorème 18.8.4 and Corollaire 18.8.8], the residual line bundle of L_K is numerically effective at every point of P^{an}_K. Choose $z \in P^{\mathrm{an}}_K$ whose

reduction is the generic point of C. By [CLD25, §17.7.9 and Lemme 17.7.10], the pair consisting of C and $\mathcal{L}|_C$ is a premodel of the residual line bundle at z; hence $\deg(\mathcal{L}|_C) \geq 0$. □

Finally we recall the notion of weakly psh functions [Wu26, Definition 6.1].

Definition 5.6. Let Y be a variety over k and $U \subseteq Y^{\mathrm{an}}$ open. A function $u\colon U \to [-\infty, \infty)$ is *weakly psh* if there is a proper birational morphism $\mu\colon Y' \to Y$ such that $u \circ \mu^{\mathrm{an}}$ is locally psh on $(\mu^{\mathrm{an}})^{-1}(U)$. Such a μ is called a *witness*.

By Chow's lemma and normalization we may take Y' normal and projective over Y. Since μ^{an} is proper and surjective, weakly psh functions are upper semicontinuous. Locally psh functions are weakly psh, and restrictions of weakly psh functions to open subsets are weakly psh.

5.2 Regularizable functions on affine varieties

In this subsection Y is a normal affine variety. Regularizable functions on Y are to psh functions on Y^{an} what psh functions of logarithmic growth are to psh functions on $\mathbf{C}^n$: they are the functions that extend, after twisting by a multiple of the boundary, to global psh functions on a projective compactification (Lemma 5.10).

Definition 5.7. (1) An *affine Fubini–Study function* on Y is a function of the form

$$g = \max\{a_1 \log|f_1| + b_1, \ldots, a_r \log|f_r| + b_r,\, b_0\},$$

where $f_1, \ldots, f_r \in k[Y]$ include a set of generators of the k-algebra $k[Y]$, $a_i \in \mathbf{Q}_{>0}$ and $b_i \in \mathbf{Q}$. We write $\mathrm{FS}(Y)$ for the set of these functions.

(2) A function u on Y^{an} has *logarithmic growth* if $u \leq g + C$ for some $g \in \mathrm{FS}(Y)$ and $C \in \mathbf{R}$.

(3) A function $u\colon Y^{\mathrm{an}} \to [-\infty, \infty)$ is *regularizable* if it is not identically $-\infty$ and is the pointwise limit of a decreasing sequence in $\mathrm{FS}(Y)$. We write $\mathcal{R}(Y)$ for the set of regularizable functions.

The class $\mathcal{R}(Y)$ is denoted $\mathrm{PSH}^{\log}_{\mathrm{reg}}(Y^{\mathrm{an}})$ in [Wu26, Definition 3.15].[2] Functions in $\mathrm{FS}(Y)$ are continuous, bounded below, and convex PL by the argument following (5.1), and their sublevel sets are compact, since they lie in closed polydiscs. Hence regularizable functions are locally psh and have logarithmic growth.

Lemma 5.8. *(1) Let $\pi\colon Y' \to Y$ be a morphism from a normal affine variety (for instance the inclusion of an affine open subset, or the identity). For $g \in \mathrm{FS}(Y)$ and $g' \in \mathrm{FS}(Y')$ there are constants C, C' with $g \circ \pi \leq Cg' + C'$.*

(2) Let $\pi\colon Y \to S$ be a finite surjective morphism to a normal affine variety S. For $g \in \mathrm{FS}(Y)$ and $h \in \mathrm{FS}(S)$ there are constants C, C' with $g \leq C\, h \circ \pi + C'$.

Proof. These are the estimates of [Wu26, Lemma 3.16].[3] The polynomial estimate in (1) applies to the finitely many pullbacks of functions defining g, expressed in the generators occurring in g'. For (2), apply the integral-equation estimate to each function defining g, and then apply (1) to its coefficients on S. Both arguments are independent of the characteristic. □

[2] The growth condition in [Wu26, Definition 3.15(2)] should read $u \leq g + O(1)$, with $g \in \mathrm{FS}(Y)$, as used in its subsequent proofs.

[3] In its proof, for $f = \sum_\alpha a_\alpha z^\alpha$, use $\log|f| \leq \max_\alpha \log|a_\alpha z^\alpha|$. For a monic relation $f^m + \sum_{i<m} b_i f^i = 0$, the root bound is $\log|f| \leq \max_{i<m}(m-i)^{-1} \log|b_i|$, with the coefficient functions pulled back to Y.

By (1), the restriction of a function of logarithmic growth to an affine open subset has logarithmic growth; we use this without further comment. Sums and finite maxima of functions of logarithmic growth have logarithmic growth, since $\mathrm{FS}(Y)$ is stable under sums and finite maxima.

Lemma 5.9. *For every $g \in \mathrm{FS}(Y)$ there are a normal projective variety P containing Y as an open subset, an ample line bundle H on P, a section $s \in H^0(P,H)$ with $P_s = Y$, a function $a \in \mathcal{H}^{\mathrm{gf}}_{\mathbf{Q}}(P,H)$ defined by sections without common zero, and $\kappa \in \mathbf{Q}_{>0}$, such that $g = \kappa(a - \log|s|)$ on Y^{an}.*

Proof. Apply [Wu26, Proposition 3.17(2)], absorbing its additive constant into the weights of a.[4] Normalizing the compactification preserves Y and ampleness, and gives the stated normal P. □

The next lemma relates regularizable functions to global psh functions. In (2), decreasing convex PL presentations near compact subsets suffice; this is the form needed in Lemma 5.16.

Lemma 5.10. *Let P be a normal projective variety, H an ample line bundle on P and $s \in H^0(P,H)$ nonzero, and put $Y \coloneqq P_s$.*

(1) Let $c \in \mathbf{Q}_{>0}$ and $\Phi \in \mathrm{PSH}(P, cH)$. There are $u_j \in \mathrm{FS}(Y)$ decreasing to $\Phi - c\log|s|$ on Y^{an} such that each $u_j + c\log|s|$ is bounded. In particular $(\Phi - c\log|s|)|_{Y^{\mathrm{an}}} \in \mathcal{R}(Y)$.

(2) Let $u\colon Y^{\mathrm{an}} \to [-\infty,\infty)$ be a function, not identically $-\infty$, such that $u \le -\kappa\log|s| + C$ for some $\kappa \in \mathbf{Q}_{>0}$ and $C \in \mathbf{R}$. Assume that every compact subset of Y^{an} has an open neighbourhood on which u is the limit of a decreasing sequence of convex PL functions. Then for every rational $c > \kappa$, the function equal to $u + c\log|s|$ on Y^{an} and to $-\infty$ on $P^{\mathrm{an}} \setminus Y^{\mathrm{an}}$ belongs to $\mathrm{PSH}(P, cH)$.

Proof. (1) Apply [Wu26, Corollary 3.18 and Proposition 3.17(1)].[5] The fixed-class regularization used there is available by [BJ22, Corollary 12.18(iii) and Example 3.9]: take $\Phi_j \in \mathcal{H}^{\mathrm{gf}}_{\mathbf{Q}}(P, cH)$ defined by sections without common zero, decreasing to Φ, and put $u_j = \Phi_j - c\log|s|$. Then $u_j + c\log|s| = \Phi_j$ is bounded.

(2) Use the truncation and gluing argument of [Wu26, Lemmas 3.25(1) and 3.26], with Lemma 5.5 as the positivity input. Fix $C_1 \in \mathbf{Q}$. Since $c > \kappa$, choose t_0 so that $u + c\log|s| \le C_1 - 2$ where $\log|s| \le t_0$. Take convex PL approximants $u_j \downarrow u$ near the compact set $\{\log|s| \ge t_0 - 1\}$. On the compact collar $t_0 - 1 \le \log|s| \le t_0$, one has $u_j + c\log|s| < C_1 - 1$ for all large j.[6] Thus $\max\{u_j + c\log|s|, C_1\}$ glues across the collar to the constant C_1 near the boundary of Y. The glued function is locally cH-convex, hence cH-psh. First let $j \to \infty$, then $C_1 \to -\infty$, using PSH2 and finiteness of u somewhere. □

We use the following closure properties.

Lemma 5.11. *(1) Every $u \in \mathcal{R}(Y)$ is finite on Y^{div}.*

(2) If $u_i \in \mathcal{R}(Y)$ decrease to a function $u \not\equiv -\infty$, then $u \in \mathcal{R}(Y)$.

[4] The displayed sections in that proof are a generating family, not necessarily a basis; this suffices to define the Fubini–Study metric.

[5] The limit in the proof of Corollary 3.18 is $g + \varphi$. In Proposition 3.17(1), replace the constant 0 in the maximum by a lower bound $-C_j$. To include generators $z = \sigma/s^N$ of $k[Y]$, add the dominated terms $\epsilon\log|z| - C_j$ with $0 < \epsilon \le c/N$; here $u_j \ge -c\log|s| - C_j$ and $\log|z| \le -N\log|s|$.

[6] This step uses compactness: the open sets where the strict inequality holds increase and cover the collar. Continuity of the limit, as required by the cited use of Dini's theorem in [Wu26, Lemma 3.25(1)], is unnecessary.

(3) *Let* $f_1, \dots, f_r \in k[Y]$, $a_i \in \mathbf{Q}_{>0}$ *and* $b_i \in \mathbf{Q}$. *If* $q \coloneqq \max_i(a_i \log|f_i| + b_i)$ *is not identically* $-\infty$, *then* $q \in \mathcal{R}(Y)$. *The class* $\mathcal{R}(Y)$ *is stable under finite maxima, finite sums, multiplication by positive rational numbers and addition of real constants.*

(4) *Let* $\pi\colon Y' \to Y$ *be a morphism from a normal affine variety. For* $u \in \mathcal{R}(Y)$, *the function* $u \circ \pi$ *is either identically* $-\infty$ *or in* $\mathcal{R}(Y')$. *In particular, restriction to affine open subsets preserves regularizability.*

(5) *Let* $u \in \mathcal{R}(Y)$ *and* $g \in \mathrm{FS}(Y)$ *with* $u \le g + C$. *For every rational* $c > 1$ *there are* $u_j \in \mathrm{FS}(Y)$ *decreasing to* u *such that each* $u_j - cg$ *is bounded.*

(6) *Let* $v\colon Y^{\mathrm{an}} \to [-\infty, \infty)$ *be a function of logarithmic growth, not identically* $-\infty$, *such that every compact subset of* Y^{an} *has an open neighbourhood on which* v *is the limit of a decreasing sequence of convex PL functions. Then* $v \in \mathcal{R}(Y)$.

Proof. Let v be as in (6), or $v \in \mathcal{R}(Y)$. Fix $g \in \mathrm{FS}(Y)$ with $v \le g + C$ and a presentation $g = \kappa(a - \log|s|)$ as in Lemma 5.9. Since a is bounded, $v \le -\kappa \log|s| + C'$, so Lemma 5.10(2) shows that $v + c\log|s| \in \mathrm{PSH}(P, cH)$ for rational $c > \kappa$. This and Lemma 5.10(1) prove (6). The positivity of the glued truncations in this argument is supplied by Lemma 5.5. Thus (6) applies to the local convex PL presentations in its statement; [Wu26, Lemma 3.26], which assumes a single tropicalization and logarithmic growth, does not by itself give this form.

(1) Psh functions are finite on $P^{\mathrm{div}} = Y^{\mathrm{div}}$, and so is $\log|s|$. (2) and (5) are the compactification arguments of [Wu26, Lemma 3.25(3), (2)], using Lemma 5.10 and PSH2. In (2), a growth bound for u_1 controls the whole sequence; in (5), use the class $c\kappa H$ in the presentation $g = \kappa(a - \log|s|)$. (3) Let $b \coloneqq \max\{0, \log|z_1|, \dots, \log|z_n|\}$ for generators z_l of $k[Y]$. The functions $\max\{q, b/j - j\}$ lie in $\mathrm{FS}(Y)$ and decrease to q because $b \ge 0$. Maxima and sums of approximants are approximants of maxima and sums, since $a\log|f| + a'\log|f'| = N^{-1}\log|f^{Na}f'^{Na'}|$; the limits are not identically $-\infty$ by (1). For a real translate $u + c$, use approximants $u_j + c_j$ with rational $c_j \downarrow c$. (4) If $u_j \in \mathrm{FS}(Y)$ decrease to u and $b' \ge 0$ is as in (3) for Y', then $\max\{u_j \circ \pi, b'/j - j\} \in \mathrm{FS}(Y')$ decrease to $u \circ \pi$. □

Lemma 5.12. *Let* $0 \ne f \in k[Y]$ *and* $u \in \mathcal{R}(Y_f)$, *and assume* $u \le g + C$ *on* Y_f^{an} *for some* $g \in \mathrm{FS}(Y)$. *For every* $c \in \mathbf{Q}_{>0}$, *the function equal to* $u + c\log|f|$ *on* Y_f^{an} *and to* $-\infty$ *on* $V(f)^{\mathrm{an}}$ *belongs to* $\mathcal{R}(Y)$.

Proof. The truncation proof of [Wu26, Proposition 3.30] applies with Lemma 5.11(5),(6),(2).[7] Its truncations extend as convex PL functions of logarithmic growth on Y, and decrease to the asserted extension. The limit is finite at v_{triv} by Lemma 5.11(1). No characteristic assumption is used. □

The following lemma replaces [Wu26, Proposition 6.11]. Its proof there uses norms of metrics [BE21], which are established under geometric reducedness assumptions, and a flattening argument. We use instead the field-norm construction of Section 4.

Lemma 5.13. *Let* $\pi\colon Y \to S$ *be a finite flat morphism of normal affine varieties, of degree* d. *For* $s \in S^{\mathrm{an}}$, *write* $\mathcal{H}(s) \otimes_{k[S]} k[Y] = \prod_y A_y$ *as a product of local Artinian* $\mathcal{H}(s)$*-algebras, indexed by the points* $y \in Y^{\mathrm{an}}$ *with* $\pi(y) = s$, *and put* $m_y(\pi) \coloneqq \dim_{\mathcal{H}(s)} A_y$, *so that* $\sum_{\pi(y)=s} m_y(\pi) = d$. *For a function* u *on* Y^{an} *let*

$$\mathrm{N}_\pi\, u(s) \coloneqq \sum_{\pi(y)=s} m_y(\pi)\, u(y).$$

[7] In that proof, read u_ϵ as $g_\epsilon = \max\{g, -\epsilon\log|f|\}$, with $\epsilon = c/3$, and retain the constant in $u_j \le 2g_\epsilon + C_j$. If $h = \max\{2g + c\log|f|, (c/3)\log|f|\}$, the neighbourhood on which the truncation at $-j$ is constant is $\{h + C_j < -j\}$; it may depend on j.

Then N_π *maps* $\mathcal{R}(Y)$ *to* $\mathcal{R}(S)$*. If* F/k *is a complete extension,* r *denotes the natural maps from* F*-analytic spaces to* k*-analytic spaces, and* N_{π_F} *is defined by the same formula, then* $(\mathrm{N}_\pi\, u)\circ r = \mathrm{N}_{\pi_F}(u\circ r)$ *for* $u\in\mathcal{R}(Y)$*.*

Proof. Let $u = N^{-1}\max_j\{\log|b_j|+\lambda_j\}\in\mathrm{FS}(Y)$, with $b_j\in k[Y]$ (the constant term corresponds to $b_j = 1$). Since $k[Y]$ is a projective $k[S]$-module of rank d, the determinant of multiplication by $\sum_j T_j b_j$ on $k[Y][T]$, where $T=(T_j)$ are indeterminates, is a form $\sum_{|\alpha|=d} c_\alpha T^\alpha$ with $c_\alpha\in k[S]$. We claim that

$$N\,\mathrm{N}_\pi\, u = \max_{|\alpha|=d}\Big\{\log|c_\alpha| + \sum_j \alpha_j\lambda_j\Big\} \qquad \text{on } S^{\mathrm{an}}.$$

At a point s, this is the computation in the proof of Proposition 4.1, with the finite $\mathcal{H}(s)$-algebra $\prod_y A_y$ in place of $E\otimes_K\mathcal{H}(v)$: after base change to a complete algebraically closed field, the determinant on a local Artinian factor of length ℓ is the ℓ-th power of the residue of the linear form, and the weighted Gauss norm is multiplicative. No separability or reducedness of the fibres is needed. The right-hand side of the claim lies in $\mathcal{R}(S)$ by Lemma 5.11(3).

For general $u\in\mathcal{R}(Y)$ with approximants $u_i\in\mathrm{FS}(Y)$, the functions $\mathrm{N}_\pi\, u_i$ decrease to $\mathrm{N}_\pi\, u$. The only point of Y^{an} over the trivial valuation of $k(S)$ is the trivial valuation of $k(Y)$, with multiplicity d; hence $\mathrm{N}_\pi\, u(v_{\mathrm{triv}}) = du(v_{\mathrm{triv}})$ is finite by Lemma 5.11(1), and Lemma 5.11(2) gives $\mathrm{N}_\pi\, u\in\mathcal{R}(S)$. Finally, the claim for π_F involves the same coefficients c_α, which gives the compatibility with base change for $u\in\mathrm{FS}(Y)$, and hence for $u\in\mathcal{R}(Y)$ by passing to decreasing limits. □

5.3 Polynomial convexity

In this subsection Y is again a normal affine variety. Polynomial hulls replace the pseudoconvexity arguments of complex analysis: the hull inequality for regularizable functions below is the analogue of the maximum principle, and the Runge property of sublevel sets is what allows us to recover a function from its Hartogs radius in Lemma 5.16. For a compact subset $K\subseteq Y^{\mathrm{an}}$, its *polynomial hull* is

$$\widehat{K} := \{y\in Y^{\mathrm{an}} : |f(y)|\le \sup_K|f| \text{ for all } f\in k[Y]\}.$$

We call K *polynomially convex* if $\widehat{K}=K$, and an open subset $D\subseteq Y^{\mathrm{an}}$ is *Runge* if $\widehat{K}\subseteq D$ for every compact $K\subseteq D$. The hull $\widehat{K}$ is compact, being closed and contained in a closed polydisc. We say that a function u on Y^{an} satisfies the *hull inequality* if $u(y)\le\sup_K u$ for every compact $K\subseteq Y^{\mathrm{an}}$ and every $y\in\widehat{K}$. Every $u\in\mathrm{FS}(Y)$ satisfies it, since each of its terms does. So does every $u\in\mathcal{R}(Y)$: if $u_j\in\mathrm{FS}(Y)$ decrease to u, then $u(y)\le u_j(y)\le\sup_K u_j$, and $\sup_K u_j$ decreases to $\sup_K u$ because K is compact and the u_j are upper semicontinuous. An increasing union of Runge open subsets is Runge, because a compact subset of the union lies in one of them.

We use the following strict-sublevel form of [Wu26, Proposition 5.4].[8]

Lemma 5.14. *(1) Let* $H\subseteq Y^{\mathrm{an}}$ *be a polynomially convex compact subset and* A *an open neighbourhood of* H*. There is* $g\in\mathrm{FS}(Y)$ *with* $g\ge 0$*,* $H\subseteq\mathrm{int}\{g=0\}$ *and* $\{g\le 1\}\subseteq A$*.*

(2) Every Runge open subset $D\subseteq Y^{\mathrm{an}}$ *is the union of an increasing sequence of open subsets* $\{g_j<1\}$ *with* $g_j\in\mathrm{FS}(Y)$*.*

[8] Throughout the arguments from [Wu26, §5], use strict polynomial inequalities rather than identifying $\mathrm{int}\{g\le t\}$ with $\{g<t\}$. These sets can differ: $\{|T-1|<1\}\subseteq\{|T|\le 1\}$ contains the rigid point $T=1$. The separation argument below gives the required strict inequalities directly.

Proof. (1) Let $z_1, \dots, z_n$ generate $k[Y]$ and choose $R \in \mathbf{Q}$ with $H \subseteq \{\max_l \log|z_l| < R\}$. The set $E := \{\max_l \log|z_l| \le R+1\} \setminus A$ is compact. For $y \in E$ we have $y \notin \widehat{H}$, so, replacing a function by a power, there are $f_y \in k[Y]$ and $a_y \in \mathbf{Q}$ with $\sup_H \log|f_y| < a_y < \log|f_y(y)| - 1$. Finitely many of the open sets $\{\log|f_y| > a_y + 1\}$, say for $y_1, \dots, y_p$, cover E. Put

$$g := \max\Big\{0,\ \max_l(\log|z_l| - R),\ \max_\ell(\log|f_{y_\ell}| - a_{y_\ell})\Big\}.$$

All terms except 0 are negative on the open set $\{\max_l \log|z_l| < R\} \cap \bigcap_\ell \{\log|f_{y_\ell}| < a_{y_\ell}\}$, which contains H, so $g = 0$ there. If $y \notin A$, then either $\max_l \log|z_l|(y) > R+1$ or $y \in E$, and in both cases $g(y) > 1$.

(2) Since Y^{an} is second countable and locally compact, D is the union of an increasing sequence of compact sets K_j. Put $Q_0 := \varnothing$ and inductively let g_j be given by (1) for the polynomially convex compact set $H_j := \widehat{K_j \cup Q_{j-1}}$, which lies in D because D is Runge, and for $A = D$; put $Q_j := \{g_j \le 1\}$. Then $K_j \cup Q_{j-1} \subseteq H_j \subseteq \{g_j < 1\} \subseteq Q_j \subseteq D$, which gives the assertion. □

We next use Rossi's local maximum principle.

Lemma 5.15. *Let $K \subseteq Y^{\mathrm{an}}$ be compact, $x \in \widehat{K}$, and V a relatively compact open neighbourhood of x with $\overline{V} \cap K = \varnothing$. Then x lies in the hull of $\partial V \cap \widehat{K}$.*

Proof. Apply the proof of [Wu26, Theorem 5.6], using Lemma 5.14(1) for its neighbourhood construction and choosing the scaling constant rational. The pasted maximum in that proof is convex PL and agrees with a function in $\mathrm{FS}(Y)$ outside a compact set. Hence it belongs to $\mathcal{R}(Y)$ by Lemma 5.11(6), and the hull inequality gives the contradiction directly. These inputs apply in every characteristic. □

Let $\widetilde{Y} := Y \times \mathbf{A}^1$, with fibre coordinate w and projection $p\colon \widetilde{Y} \to Y$. The Hartogs radius will recover regularizability from the Runge property of $\{u \circ p + \log|w| < 0\}$. For $y \in Y^{\mathrm{an}}$ the fibre $p^{-1}(y) \subseteq \widetilde{Y}^{\mathrm{an}}$ is identified with the Berkovich affine line over $\mathcal{H}(y)$, and $(y, 0)$ denotes its origin. For an open subset $D \subseteq \widetilde{Y}^{\mathrm{an}}$ containing the zero section $\{(y,0) : y \in Y^{\mathrm{an}}\}$, let $d_D(y) \in (0, +\infty]$ be the supremum of the radii r such that the closed disc $\{|w| \le r\}$ of $p^{-1}(y)$ is contained in D.

Lemma 5.16. *Let $u\colon Y^{\mathrm{an}} \to [-\infty, \infty)$ be upper semicontinuous, of logarithmic growth and not identically $-\infty$. If $\{u \circ p + \log|w| < 0\} \subseteq \widetilde{Y}^{\mathrm{an}}$ is Runge, then $u \in \mathcal{R}(Y)$.*

Proof. Apply the proof of [Wu26, Proposition 5.7] to $D = \{u \circ p + \log|w| < 0\}$, whose Hartogs radius is $d_D = \mathrm{e}^{-u}$. By Lemma 5.14(2), exhaust D by increasing strict polyhedra $W_j = \{g_j < 1\} \cap \{\log|w| < j\}$, and put $\Omega_j = \{y : (y,0) \in W_j\}$. The fibrewise calculation of [Wu26, Lemma 5.8], made with these strict inequalities, shows that $-\log d_{W_j}$ is convex PL on Ω_j.[9] Compact fibre discs show that these radii increase to d_D. Since $\Omega_j \uparrow Y^{\mathrm{an}}$, the resulting decreasing approximation exists on a neighbourhood of every compact subset of Y^{an}. Now apply Lemma 5.11(6), as in Wu's argument. □

The following lemma is [Wu26, Proposition 5.10]. The proof given there rests on an incorrect ideal-theoretic claim [Wu26, Lemma 5.11]; we give a direct argument.

Lemma 5.17. *Let $0 \neq f \in k[Y]$, let $K \subseteq Y^{\mathrm{an}}$ be compact, and let $x \in \widehat{K} \cap Y_f^{\mathrm{an}}$, where $\widehat{K}$ is the hull in Y. If $v \in \mathcal{R}(Y_f)$ satisfies $v \le g + C$ for some $g \in \mathrm{FS}(Y)$, then $v(x) \le \sup_{K \cap Y_f^{\mathrm{an}}} v$.*

[9] If the defining inequalities are $\log|\sum_\ell a_{\alpha\ell}(y) w^\ell| < c_\alpha$, on Ω_j the Gauss norm gives $-\log d_{W_j} = \max\{-j, M_j\}$, where $M_j = \max_{\alpha, \ell \ge 1}(\log|a_{\alpha\ell}| - c_\alpha)/\ell$. The constant term inequalities hold there by definition, and coefficients vanishing at a support are locally dominated by $-j$.

Proof. Choose $M \geq 0$ with $\log|f| \leq M$ on K and $|\log|f(x)|| \leq M$. The set $K \cap Y_f^{\mathrm{an}}$ is nonempty, since $x \in \widehat{K}$ and $|f(x)| > 0$. For $c \in \mathbf{Q}_{>0}$, let v_c be the extension of $v + c\log|f|$ by $-\infty$. By Lemma 5.12, $v_c \in \mathcal{R}(Y)$, so the hull inequality gives

$$v(x) - cM \leq v_c(x) \leq \sup_K v_c \leq \sup_{K \cap Y_f^{\mathrm{an}}} v + cM.$$

The last supremum is bounded above by $\sup_K g + C$. If it is finite, let $c \downarrow 0$; if it is $-\infty$, the same inequality gives $v(x) = -\infty$. □

5.4 A strictly psh exhaustion and gluing

In this subsection Y is a normal affine variety. We need a regularizable function that admits supporting functions with a strict drop outside prescribed neighbourhoods, as in [Wu26, Lemmas 6.6 and 6.7]. We construct it using the countability of k and a series of the functions $\log(1+|f|)$, $f \in k[Y]$, whose supporting lines are explicit.

Let $\mathbf{T} \coloneqq \mathbf{R} \cup \{+\infty\}$ with its natural topology, and let $\lambda\colon \mathbf{T} \to [0,\infty)$ be $\lambda(\tau) \coloneqq \log(1+\mathrm{e}^{-\tau})$, with $\lambda(+\infty) = 0$. Thus $\lambda(-\log|f|) = \log(1+|f|)$ on Y^{an} for $f \in k[Y]$.

Lemma 5.18. *There is $\rho \in \mathcal{R}(Y)$ with $\rho \geq 0$ such that for every $x \in Y^{\mathrm{an}}$ and every open neighbourhood V of x there are $\rho' \in \mathcal{R}(Y)$ and $\delta > 0$ with*

$$\rho' \leq \rho, \qquad \rho'(x) = \rho(x), \qquad \rho' \leq \rho - \delta \text{ on } Y^{\mathrm{an}} \setminus V.$$

Proof. Supporting lines. For $\tau^0 \in \mathbf{R}$ let $p \coloneqq \mathrm{e}^{-\tau^0}/(1+\mathrm{e}^{-\tau^0}) \in (0,1)$ and $\ell_{\tau^0}(\tau) \coloneqq \lambda(\tau^0) - p(\tau - \tau^0) \in [-\infty,\infty)$; for $\tau^0 = +\infty$ let $\ell_{\tau^0} \coloneqq 0$. We claim that $\lambda - \ell_{\tau^0} \geq 0$ on $\mathbf{T}$, with equality only at τ^0, and that it is bounded below by a positive constant outside every neighbourhood of τ^0. For $\tau^0 = +\infty$ this holds because λ is positive and decreasing on $\mathbf{R}$. For $\tau^0 \in \mathbf{R}$, the function $\lambda - \ell_{\tau^0}$ is strictly convex on $\mathbf{R}$ with vanishing derivative at τ^0, it tends to $+\infty$ at both ends of $\mathbf{R}$, since $\lambda(\tau) \geq \max\{0, -\tau\}$ and $p < 1$, and it equals $+\infty$ at $+\infty$.

Regularizability. Let $f \in k[Y]$ and $x \in Y^{\mathrm{an}}$. We claim that $\log(1+|f|) \in \mathcal{R}(Y)$, and that $\ell_{\tau^0}(-\log|f|) \in \mathcal{R}(Y)$ for $\tau^0 \coloneqq -\log|f|(x)$. For the first claim, the binomial formula applied to the nonnegative real number $|f|$ gives $P_m \leq \log(1+|f|) \leq P_m + m^{-1}\log(m+1)$, where

$$P_m \coloneqq \frac{1}{m}\max\left\{0,\ \max_{1\leq\alpha\leq m}\left(\log\binom{m}{\alpha} + \log|f^\alpha|\right)\right\}.$$

Rounding the constants of $P_m + m^{-1}\log(m+1)$ up to rational numbers, for m large, gives functions Q_j of the form in Lemma 5.11(3) with $\log(1+|f|) \leq Q_j \leq \log(1+|f|) + 2^{-j-2}$. The functions $Q_j + 2^{-j}$ are regularizable and decrease to $\log(1+|f|) \geq 0$, so Lemma 5.11(2) applies. For the second claim we may assume τ^0 finite; then $f \neq 0$ and $\ell_{\tau^0}(-\log|f|) = C + p\log|f|$ with $C \coloneqq \lambda(\tau^0) + p\tau^0$. Fix generators $z_1,\dots,z_n$ of $k[Y]$ and put $b \coloneqq 1 + \max\{0, \log|z_1|, \dots, \log|z_n|\} \in \mathrm{FS}(Y)$. Choose rational numbers $a_m \uparrow p$, $a'_m \downarrow p$ and $C_m \downarrow C$ with $a_m > 0$. The functions

$$q_m \coloneqq \max\{C_m + a_m\log|f|,\ C_m + a'_m\log|f|,\ b/m - m\}$$

lie in $\mathrm{FS}(Y)$. They decrease, because $\max\{a\tau, a'\tau\}$ decreases as the interval $[a,a']$ shrinks, whatever the sign of τ, and their limit is $C + p\log|f|$, which is finite at v_{triv}.

Construction of ρ. Enumerate the countable set $k[Y]$ as $f_1, f_2, \dots$, and put $h_j \coloneqq \log(1+|f_j|)$. By the degree estimate of Lemma 5.8(1) there are integers $C_j \geq 1$ with $0 \leq h_j \leq C_j b$. Put $\gamma_j \coloneqq 2^{-j}/C_j$ and $\rho \coloneqq \sum_j \gamma_j h_j$, so that $0 \leq \rho \leq b$. The regularizable functions $R_N \coloneqq \sum_{j\leq N}\gamma_j h_j + 2^{-N}b$ decrease to ρ, because $R_{N+1} - R_N = \gamma_{N+1}h_{N+1} - 2^{-N-1}b \leq 0$; hence $\rho \in \mathcal{R}(Y)$.

Strictness. Let $x \in Y^{\mathrm{an}}$ and V an open neighbourhood of x. Since the topology of Y^{an} is the coarsest one making all $|f|$, $f \in k[Y]$, continuous, there are a nonempty finite set J of indices and open neighbourhoods U_j of $\tau_j := -\log|f_j|(x)$ in $\mathbf{T}$ such that $\{y : -\log|f_j|(y) \in U_j \text{ for all } j \in J\} \subseteq V$. Let $\delta_j > 0$ be a lower bound of $\lambda - \ell_{\tau_j}$ outside U_j, and put

$$\rho' := \rho - \sum_{j\in J} \gamma_j\,(\lambda - \ell_{\tau_j})(-\log|f_j|).$$

Then $\rho - \rho'$ is nonnegative, vanishes at x, and is at least $\delta := \min_{j\in J}\gamma_j\delta_j$ outside V. Finally, for $N \geq \max J$ the regularizable functions $R_N - \sum_{j\in J}\gamma_j(h_j - \ell_{\tau_j}(-\log|f_j|))$ decrease to ρ', which is finite at x; so $\rho' \in \mathcal{R}(Y)$. □

The Runge property needed in Lemma 5.16 will be obtained from hull inequalities on $\widetilde{Y} = Y \times \mathbf{A}^1$, as follows. Recall that $p\colon \widetilde{Y} \to Y$ is the projection and w the coordinate on $\mathbf{A}^1$.

Lemma 5.19. *Let $u\colon Y^{\mathrm{an}} \to [-\infty, \infty)$ be upper semicontinuous, of logarithmic growth and not identically $-\infty$. Put $\widetilde{u} := u \circ p + \log|w|$, and let $\widetilde{\rho}$ be a function as in Lemma 5.18 for $\widetilde{Y}$. If $\widetilde{u} + c\widetilde{\rho}$ satisfies the hull inequality for every rational $c > 0$, then $u \in \mathcal{R}(Y)$.*

Proof. Let $c > 0$ be rational. If $K \subseteq \{\widetilde{u} + c\widetilde{\rho} < 0\}$ is compact, then $\sup_K(\widetilde{u} + c\widetilde{\rho}) < 0$ by upper semicontinuity, so $\widehat{K} \subseteq \{\widetilde{u} + c\widetilde{\rho} < 0\}$ by the hull inequality. Thus these open sets are Runge. As $c \downarrow 0$ they increase to $\{\widetilde{u} < 0\}$, since $\widetilde{\rho}$ is finite and nonnegative. Hence $\{\widetilde{u} < 0\}$ is Runge, and Lemma 5.16 gives $u \in \mathcal{R}(Y)$. □

The following gluing statement is [Wu26, Lemma 6.8 and Corollary 6.9], with the local regularizability hypothesis used in their proofs.[10]

Proposition 5.20. *Let $Z \subsetneq Y$ be closed (possibly empty) and u a weakly psh function on Y^{an} of logarithmic growth. Assume that $Y \setminus Z$ is covered by affine open subsets U_α with $u|_{U_\alpha^{\mathrm{an}}} \in \mathcal{R}(U_\alpha)$, and let ρ be as in Lemma 5.18.*

(1) For every rational $c > 0$, every compact $K \subseteq Y^{\mathrm{an}}$ and every $x \in \widehat{K} \setminus Z^{\mathrm{an}}$, we have $(u + c\rho)(x) \leq \sup_K(u + c\rho)$.

(2) If $Z = \varnothing$, then $u \in \mathcal{R}(Y)$.

Proof. (1) Use the proof of [Wu26, Lemma 6.8], with Lemmas 5.15, 5.17 and 5.18 as its inputs. In the notation of that proof, the auxiliary function $u + c\rho' + \epsilon v$ is regularizable on the principal neighbourhood $Y_g \subseteq U_\alpha$ by Lemma 5.11(3),(4), and has logarithmic growth on Y since $\rho' \leq \rho$. Thus its strict boundary drop contradicts the principal-open hull inequality exactly as there. The proof uses only these inputs, which hold over k.

(2) Apply (1) with $Z = \varnothing$ to $\widetilde{Y} = Y \times \mathbf{A}^1$ and $\widetilde{u} = u \circ p + \log|w|$, as in [Wu26, Corollary 6.9]. A witness for u pulls back to a witness on $\widetilde{Y}$, and Lemma 5.11(3),(4) gives regularizability on each $U_\alpha \times \mathbf{A}^1$. Then Lemma 5.19 applies; u is not identically $-\infty$, being regularizable on some U_α. □

5.5 Norms along generically étale projections

To extend regularizability across a closed subset, we project a neighbourhood of one of its points finitely onto an open subset of $\mathbf{A}^n$ and push functions forward by a weighted sum over

[10] In Lemma 6.8, assume $u|_{U_\alpha^{\mathrm{an}}} \in \mathcal{R}(U_\alpha)$, as in the statement below; regularizability of $u + c\rho$ alone does not justify the auxiliary function $u + c\rho' + \epsilon v$. One may take $v = \max_i \log|f_i|$ for generators of the ideal of Z, and choose ϵ rational.

the fibres. Outside a dense open subset over which the projection is étale, the weights are defined by restricting to general lines, and regularity of the pushforward is checked on curves. We use subharmonic functions on analytic curves over an arbitrary complete non-Archimedean field in the sense of [GR25, Definition 3.4], which may be identically $-\infty$ on a component. By Lemma 5.2(4), the pullback of a locally psh function under a morphism from a curve is subharmonic.

Lemma 5.21. *Let F be a complete non-Archimedean field.*

(1) Let $q\colon C' \to C$ be a finite flat surjective morphism of normal algebraic F-curves, with C integral. For $y \in C^{\mathrm{an}}$ write $\mathcal{H}(y) \otimes_{\mathcal{O}_C} q_*\mathcal{O}_{C'} = \prod_x A_x$, the product of local Artinian algebras indexed by the points $x \in C'^{\mathrm{an}}$ over y, and put $m_x(q) \coloneqq \dim_{\mathcal{H}(y)} A_x$. If u is subharmonic on C'^{an}, then $\mathrm{N}_q\, u(y) \coloneqq \sum_{q(x)=y} m_x(q)u(x)$ is subharmonic on C^{an}.

(2) Two subharmonic functions on the analytification of an algebraic F-curve that agree outside finitely many rigid points are equal.

(3) Assume that F is an algebraically closed complete extension of k, let r denote the natural maps from F-analytic spaces to k-analytic spaces, and let u be a weakly psh function on Y^{an} for a variety Y over k. For every morphism $p\colon C \to Y_F$ from a smooth F-curve, $u \circ r \circ p^{\mathrm{an}}$ is subharmonic on C^{an}.

Proof. (1) Let $K = F(C)$, let E_i be the function fields of the components C'_i of C', and let ν_i be the inseparable degree of E_i/K. Choose a finite normal extension L/K containing a conjugate of each E_i, and let $r_L\colon D \to C$ be the normalization of C in L, which is finite. Every K-embedding $\sigma\colon E_i \to L$ induces a C-morphism $p_{i,\sigma}\colon D \to C'_i$, since C'_i is the normalization of C in E_i. We claim that

$$(\mathrm{N}_q\, u) \circ r_L^{\mathrm{an}} = \sum_i \nu_i \sum_{\sigma\colon E_i \hookrightarrow L} u \circ p_{i,\sigma}^{\mathrm{an}} \tag{5.2}$$

for every function u. Work over an affine open $\operatorname{Spec} R \subseteq C$ over which $q_*\mathcal{O}_{C'}$ is free with basis $b_1, \dots, b_d$, where d is the degree of q, and let S be the integral closure of R in L. Since the generic fibre of q decomposes into the fields E_i, and the characteristic polynomial of an element of E_i over K is the product of its conjugates, each with multiplicity ν_i, we have

$$\det(Z - m_{b(T)}) = \prod_i \prod_{\sigma\colon E_i \hookrightarrow L} \left(Z - \sigma(b(T))\right)^{\nu_i} \qquad \text{in } S[T_1, \dots, T_d, Z],$$

where $b(T) = \sum_j T_j b_j$ and $m_{b(T)}$ is multiplication by $b(T)$. Fix $z \in D^{\mathrm{an}}$ and put $y = r_L(z)$. Specialize to $\mathcal{H}(z)$ and substitute for T the coordinates of the idempotent of A_x. The left side becomes $Z^{d-m_x(q)}(Z-1)^{m_x(q)}$, and a factor on the right becomes $Z-1$ exactly when $p_{i,\sigma}(z) = x$. Hence $m_x(q) = \sum_{(i,\sigma)\colon p_{i,\sigma}(z)=x} \nu_i$, which proves (5.2). Its right side is subharmonic by [GR25, Proposition 3.5(4), (8)]. Since r_L^{an} is closed and surjective, $\mathrm{N}_q\, u$ is upper semicontinuous, and it is subharmonic by finite surjective descent [GR25, Proposition 3.5(8)].

(2) Let h be a subharmonic function and z a rigid point. By upper semicontinuity $h(z) \geq \limsup_{x \to z,\, x \neq z} h(x)$. If the inequality were strict, h would have a strict local maximum at z, contradicting the maximum principle [GR25, Proposition 3.5(6)]: the analytification of an algebraic curve has no boundary and no isolated points. Hence $h(z)$ is determined by the values of h near z, also when $h(z) = -\infty$.

(3) Subharmonicity being local [GR25, Proposition 3.5(1)], we may assume C connected, hence irreducible. Let $\mu\colon Y' \to Y$ be a witness. Then $u \circ r \circ \mu_F^{\mathrm{an}}$ is locally psh on $Y_F'^{\mathrm{an}}$ by Lemma 5.2(2). The fibre product $C \times_{Y_F} Y'_F$ is proper and surjective over C; choose an irreducible component dominating C, a closed point of its generic fibre over C, and let C'' be the normalization of its closure. Then C'' is a smooth curve, finite and surjective over C,

with a morphism $C'' \to Y'_F$ lifting p. The pullback of $u \circ r \circ p^{\mathrm{an}}$ to C''^{an} is the pullback of $u \circ r \circ \mu_F^{\mathrm{an}}$, hence subharmonic, and finite surjective descent [GR25, Proposition 3.5(8)] shows that $u \circ r \circ p^{\mathrm{an}}$ is subharmonic. This works even if $p(C)$ lies in the locus over which μ is not an isomorphism. □

The next proposition is [Wu26, Proposition 6.12], extended to positive characteristic. The proof in [Wu26] applies a regularization theorem for smooth affine curves over the completed residue field $\mathcal{H}(s)$ of a point s, which need not be discretely or trivially valued; we only use subharmonicity over an algebraically closed complete extension, via Lemma 5.21. Instead of [Wu26, Proposition 6.2(2)], applied there on the étale locus, we assume regularizability on an affine open cover of $Y \setminus Z$ and take the étale locus off $\pi(Z)$.

Proposition 5.22. *Let Y be a normal affine variety of dimension $n \geq 1$, $S \subseteq \mathbf{A}^n$ a principal open subset, $\pi\colon Y \to S$ a finite surjective generically étale morphism of degree d, and $Z \subsetneq Y$ closed. There are positive integers $m_y(\pi)$, $y \in Y^{\mathrm{an}}$, with $\sum_{\pi(y)=s} m_y(\pi) = d$ for every $s \in S^{\mathrm{an}}$, such that the following holds. If u is a weakly psh function on Y^{an} of logarithmic growth, and $Y \setminus Z$ is covered by affine open subsets on which u is regularizable, then*

$$\mathrm{N}_\pi\, u(s) \coloneqq \sum_{\pi(y)=s} m_y(\pi) u(y)$$

defines an element of $\mathcal{R}(S)$.

Proof. Choice of weights. Choose $0 \neq h \in k[S]$ such that $S_0 \coloneqq S_h \subseteq S \setminus \pi(Z)$ and π is étale over S_0, and put $Y_0 \coloneqq \pi^{-1}(S_0)$, an affine open subset of $Y \setminus Z$. Over S_0^{an} use the weights of Lemma 5.13. Let $s \in S^{\mathrm{an}} \setminus S_0^{\mathrm{an}}$, let F be the completion of an algebraic closure of $\mathcal{H}(s)$, and let $s_F \in S_F(F)$ be the rigid point induced by s. Choose a line $\Lambda \subseteq \mathbf{A}^n_F$ through s_F that is not contained in the closure of $(S \setminus S_0)_F$ in $\mathbf{A}^n_F$; this is possible because the lines through s_F contained in a closed subset of dimension less than n form a proper closed subset of the space of lines through s_F. Let $C_s \coloneqq \Lambda \cap S_F$; it is a smooth affine curve, and $C_s \cap (S_0)_F$ is dense in C_s. Let C'_s be the normalization of the reduction of $Y_F \times_{S_F} C_s$. Since k is algebraically closed, Y_F is integral of dimension n. By Krull's height theorem every irreducible component of $Y_F \times_{S_F} C_s$, which is cut out in Y_F by $n-1$ equations, has dimension at least one, and finiteness over C_s gives dimension at most one; so every component is a curve dominating C_s. Over $C_s \cap (S_0)_F$ the fibre product is étale over C_s, hence reduced and normal, so it is not changed there by reduction and normalization. Thus C'_s is a smooth curve, and $q_s\colon C'_s \to C_s$ is finite, flat (being torsion free over a smooth curve) and of degree d. Every point of Y^{an} over s is the image of a point of Y_F^{an} over s_F, which in turn is the image of a rigid point of C'^{an}_s over s_F. For $y \in \pi^{-1}(s)$, let $m_y(\pi)$ be the sum of the multiplicities $m_x(q_s)$ over the points $x \in q_s^{-1}(s_F)$ mapping to y. Then $m_y(\pi) \geq 1$ and $\sum_y m_y(\pi) = d$. These choices do not depend on u.

Regularity. Let u be as in the statement. The affine open subset Y_0 of $Y \setminus Z$ is covered by the intersections of Y_0 with the given affine open subsets, which are affine and on which u is regularizable by Lemma 5.11(4); so $u|_{Y_0^{\mathrm{an}}} \in \mathcal{R}(Y_0)$ by Proposition 5.20(2), and $v_0 \coloneqq \mathrm{N}_{\pi|_{Y_0}} u \in \mathcal{R}(S_0)$ by Lemma 5.13. By Lemma 5.8(2), and since functions in $\mathrm{FS}(S)$ are bounded below, $u \leq B' g \circ \pi + B''$ for some $g \in \mathrm{FS}(S)$ and constants $B' \in \mathbf{Q}_{>0}$, $B'' \in \mathbf{R}$, so $v_0 \leq dB'g + dB''$ on S_0^{an}, where $dB'g \in \mathrm{FS}(S)$.

Extension to S. We extend v_0 to S using the envelope property of $\mathbf{P}^n$; this replaces [Wu26, Theorem 3.31], whose proof uses a smooth compactification with ample boundary. Write $S = \{h_1 \neq 0\}$ with $h_1 \in k[x_1, \ldots, x_n]$ of degree e, regard $\mathbf{A}^n$ as the chart $\{x_0 \neq 0\}$ of $P \coloneqq \mathbf{P}^n$, and let $\sigma \in H^0(P, H)$, $H \coloneqq \mathcal{O}_P(e+1)$, be the product of x_0 and the homogenization of h_1, so that $P_\sigma = S$. Every $f \in k[S]$ has the form τ/σ^N with $\tau \in H^0(P, NH)$, so $\log|f| \leq -N \log|\sigma|$. Fix rational numbers $c_i \downarrow 0$. By Lemma 5.12, the extension w_i of $v_0 + c_i \log|h|$ by $-\infty$ lies

in $\mathcal{R}(S)$, and by the previous remark there are $\kappa \in \mathbf{Q}_{>0}$ and $B_0 \in \mathbf{R}$, independent of i, with $w_i \le -\kappa \log|\sigma| + B_0$. Fix a rational $a > \kappa$. By Lemma 5.10(2), the extension ϕ_i of $w_i + a\log|\sigma|$ by $-\infty$ lies in $\mathrm{PSH}(P, aH)$, and $\phi_i \le B_0$. Since P is smooth, $\mathrm{EP}_P(c_1(aH))$ holds by Theorem 3.1, so $\psi_j := \sup^*_{i\ge j} \phi_i$ lies in $\mathrm{PSH}(P, aH)$. On the open set S_0^{an} we have $\sup_{i\ge j}\phi_i = v_0 + a\log|\sigma| + \max\{c_j \log|h|, 0\}$, which is upper semicontinuous there; hence ψ_j is given by the same formula on S_0^{an}. The ψ_j decrease to a function ψ equal to $v_0 + a\log|\sigma|$ on S_0^{an}, which is finite at $v_{\mathrm{triv}} \in S_0^{\mathrm{an}}$ by Lemma 5.11(1); so ψ is aH-psh by PSH2. By Lemma 5.10(1), $v := (\psi - a\log|\sigma|)|_{S^{\mathrm{an}}}$ lies in $\mathcal{R}(S)$, and $v = v_0$ on S_0^{an}.

Identification off S_0. It remains to show $v(s) = \mathrm{N}_\pi u(s)$ for $s \notin S_0^{\mathrm{an}}$. With the notation above, $v \circ r$ restricts to a locally psh, hence subharmonic, function on C_s^{an}. The pullback of $u \circ r$ to $C_s'^{\mathrm{an}}$ is subharmonic by Lemma 5.21(3), so its norm under q_s is subharmonic on C_s^{an} by Lemma 5.21(1). Over $C_s \cap (S_0)_F$, the map $C_s' \to Y_F \times_{S_F} C_s$ is an isomorphism, so the fibre algebras of q_s and of π_F agree there; by the base change compatibility in Lemma 5.13, the two subharmonic functions agree on $(C_s \cap (S_0)_F)^{\mathrm{an}}$. They differ at most at the finitely many rigid points of $C_s \setminus (S_0)_F$, hence are equal by Lemma 5.21(2). Evaluating at s_F gives $v(s) = \sum_y m_y(\pi) u(y)$. □

5.6 Extension across a closed subset

We now show that regularizability extends across proper closed subsets. We first recall how to project a neighbourhood of a smooth point of a closed subset finitely onto an open subset of $\mathbf{A}^n$. The following is [Wu26, Lemma 6.16], supplemented by generic étaleness, which is needed in positive characteristic.

Lemma 5.23. *Let Y be an affine variety of dimension $n \ge 1$, $Z \subsetneq Y$ a closed subset, and $z \in Z$ a closed point at which Z is smooth. There are an affine open neighbourhood U of z, a principal open subset $S \subseteq \mathbf{A}^n$ and a finite surjective generically étale morphism $\pi\colon U \to S$ whose restriction to $Z \cap U$ is a closed immersion.*

Proof. Use the successive general projections in the proof of [Wu26, Lemma 6.16], and shrink the target to a principal affine neighbourhood of the image of z.[11] To ensure generic étaleness in positive characteristic, fix also a smooth closed point w of the projective closure of Y. At each stage choose the projection centre so that the projection is a local isomorphism at the image of w, or unramified there at the last stage. This excludes a further proper closed subset of centres by [Stacks, Tag 0B1Q]. The resulting finite map is unramified at w, hence generically étale. □

The main step is the following variant of [Wu26, Lemma 6.15]. We follow the strategy there: at a point where the hull inequality fails, a norm along the projection produces a regularizable function on S that violates it. The barrier is built from a partition $f_1 + f_2 = 1$ separating Z from the rest of its preimage, which avoids the polyhedral neighbourhoods of [Wu26, Lemma 6.15, Step 2] and the constancy of a local degree asserted in [Wu26, Lemma 6.15, Step 3], which is not justified as stated.

Proposition 5.24. *Let Y be a normal affine variety of dimension $n \ge 1$, $S \subseteq \mathbf{A}^n$ a principal open subset, $\pi\colon Y \to S$ a finite surjective generically étale morphism, and $Z \subsetneq Y$ a closed subset such that $\pi|_Z$ is a closed immersion and $Z' := \pi^{-1}(\pi(Z)) \setminus Z$ is closed in Y. Let u be a weakly psh function on Y^{an} of logarithmic growth, and assume that $Y \setminus Z$ is covered by affine open subsets on which u is regularizable. Then $u \in \mathcal{R}(Y)$.*

[11] At the end of that proof, take $U = r^{-1}(S)$ rather than $r^{-1}(W)$. The projection argument only uses smoothness of Z at z: apply [Stacks, Tag 0B1Q] to the component through z, and avoid the cones joining z to the other components and to the boundary of the compactification.

Proof. Let ρ be as in Lemma 5.18, let d be the degree of π, and let $m_y := m_y(\pi)$ be the weights of Proposition 5.22 for π and Z. We show that $u_c := u + c\rho$ satisfies the hull inequality for every rational $c > 0$. Since $\widetilde{Y} = Y \times \mathbf{A}^1$, $Z \times \mathbf{A}^1$, $\pi \times \mathrm{id}$ and $u \circ p + \log|w|$ satisfy the same hypotheses (as in the proof of Proposition 5.20(2)), Lemma 5.19 then gives $u \in \mathcal{R}(Y)$; note that u is not identically $-\infty$, being regularizable on a nonempty open subset.

Suppose that $u_c(x_0) > \sup_L u_c$ for some compact $L \subseteq Y^{\mathrm{an}}$ and $x_0 \in \widehat{L}$. The upper semicontinuous function u_c attains its maximum over the compact set $\widehat{L}$ at some point x. Subtracting the finite constant $u_c(x)$ from u, we may assume that $u_c(x) = 0 = \max_{\widehat{L}} u_c$. Choose a rational $\delta > 0$ with $\sup_L u_c \leq -\delta$; in particular $x \notin L$. By Proposition 5.20(1), $x \in Z^{\mathrm{an}}$.

Since Z and Z' are disjoint closed subsets of the affine variety Y, there are $f_1, f_2 \in k[Y]$ with $f_1 + f_2 = 1$, $f_1 = 0$ on Z' and $f_2 = 0$ on Z; then $f_1 \neq 0$, as $Z \neq \varnothing$. By the ultrametric inequality, the open sets $O_1 := \{|f_2| < 1\} \supseteq Z^{\mathrm{an}}$ and $O_2 := \{|f_1| < 1\} \supseteq Z'^{\mathrm{an}}$ are disjoint, $|f_1| = 1$ on O_1 and $|f_2| = 1$ on O_2. The map π^{an} is proper, hence closed, and it is injective on Z^{an} since $\pi|_Z$ is a closed immersion. Hence the fibre $\pi^{-1}(\pi(x))$ consists of x and of a compact subset Γ of Z'^{an}. We choose, in this order:

(1) a relatively compact open neighbourhood A of Γ with $\overline{A} \subseteq \{\log|f_1| < -1\}$, and a rational $M \geq 0$ with $u_c \leq M$ on $\overline{A}$;

(2) an open neighbourhood $V_x \subseteq \{\log|f_2| < -1\}$ of x on which $u_c < \epsilon := \delta/(2d)$;

(3) the open neighbourhood $V := S^{\mathrm{an}} \setminus \pi(Y^{\mathrm{an}} \setminus (V_x \cup A))$ of $\pi(x)$, which satisfies $\pi^{-1}(V) \subseteq V_x \cup A$;

(4) a relatively compact open neighbourhood B of x with $\overline{B} \subseteq V_x \cap \pi^{-1}(V)$ and $\overline{B} \cap L = \varnothing$;

(5) $\rho' \in \mathcal{R}(Y)$ and $\delta' > 0$ as in Lemma 5.18 for x and B.

Put $u' := u + c\rho' \leq u_c$ and $\eta := \max\{\delta \log|f_2|, -\delta\} \in \mathcal{R}(Y)$, and

$$u^\sharp := \max\{u' + (M+1)\log|f_1|,\ \eta\}.$$

This function is weakly psh, with the witness of u, and of logarithmic growth, and it is regularizable on every affine open subset on which u is, by Lemma 5.11(3),(4). On $V_x \subseteq O_1$ we have $\log|f_1| = 0$ and $\eta = -\delta$, so $u^\sharp = \max\{u', -\delta\} < \epsilon$. On $A \subseteq O_2$ we have $\eta = 0$ and $u' + (M+1)\log|f_1| < M - (M+1)$, so $u^\sharp = 0$. By Proposition 5.22, $h := \mathrm{N}_\pi u^\sharp \in \mathcal{R}(S)$. Since $u^\sharp = 0$ on $\Gamma \subseteq A$ and $u'(x) = u_c(x) = 0$, we get $h(\pi(x)) = m_x u^\sharp(x) = 0$.

By Lemma 5.15, x lies in the hull of the compact set $K := \partial B \cap \widehat{L}$; hence $\pi(x)$ lies in the hull of $\pi(K)$ in S, and the hull inequality for h gives $0 \leq \sup_K h \circ \pi$. Since $h \circ \pi$ is upper semicontinuous, it suffices to show $h(\pi(y)) < 0$ for every $y \in K$ to reach a contradiction. We have $y \in V_x$, and $\pi^{-1}(\pi(y)) \subseteq V_x \cup A$ since $\pi(y) \in V$. As $y \in \widehat{L} \setminus B$, we have $u'(y) \leq u_c(y) - c\delta' \leq -c\delta'$. If $y \notin Z^{\mathrm{an}}$, then $u_c(y) \leq \sup_L u_c \leq -\delta$ by Proposition 5.20(1), so $u^\sharp(y) = -\delta$; since $u^\sharp \leq \epsilon$ at the other points of the fibre and $m_y \geq 1$,

$$h(\pi(y)) \leq -\delta\, m_y + (d - m_y)\,\epsilon < -\delta/2.$$

If $y \in Z^{\mathrm{an}}$, the other points of $\pi^{-1}(\pi(y))$ lie in Z'^{an}, which does not meet $V_x \subseteq O_1$; so they lie in A, where $u^\sharp = 0$, and $h(\pi(y)) = m_y \max\{u'(y), -\delta\} \leq m_y \max\{-c\delta', -\delta\} < 0$. □

The inductions in [Wu26, Proposition 6.14 and Theorem 6.3], the latter starting from the smooth locus, are replaced by the following Noetherian argument.

Theorem 5.25. *Let Y be a normal affine variety over the countable field k, $Z \subsetneq Y$ a closed subset, and u a weakly psh function on Y^{an} of logarithmic growth. If $Y \setminus Z$ is covered by affine open subsets on which u is regularizable, then $u \in \mathcal{R}(Y)$.*

Proof. Let $\Sigma \subseteq Y$ be the union of the affine open subsets U with $u|_{U^{\mathrm{an}}} \in \mathcal{R}(U)$, and $W := Y \setminus \Sigma$, a closed subset of Z. By Proposition 5.20(2) it suffices to show $W = \varnothing$. Suppose not; then $n := \dim Y \geq 1$. Since W is reduced and k is perfect, W is smooth at some closed point z, and removing from Y the singular locus of W and the irreducible components of W not containing z, we find a principal open neighbourhood Y_1 of z such that $W \cap Y_1$ is smooth and irreducible. Lemma 5.23 gives an affine open neighbourhood $U \subseteq Y_1$ of z and a finite surjective generically étale morphism $\pi\colon U \to S$ onto a principal open subset of $\mathbf{A}^n$ that restricts to a closed immersion on the irreducible set $W_U := W \cap U$. Put $Z' := \pi^{-1}(\pi(W_U)) \setminus W_U$, and let $\overline{Z'}$ be its closure in U, so that $\overline{Z'} = Z' \cup E$ with $E := \overline{Z'} \cap W_U$. Every irreducible component of $\overline{Z'}$ meets Z', so it is not contained in W_U, and it has dimension at most $\dim \pi^{-1}(\pi(W_U)) = \dim W_U$; hence it does not contain W_U. Thus E is a proper closed subset of W_U, and $\pi(E)$ is a proper closed subset of the irreducible set $\pi(W_U)$. Choose $g \in k[S]$ vanishing on $\pi(E)$ but not on $\pi(W_U)$, and put $S_1 := S_g$ and $U_1 := \pi^{-1}(S_1)$. Then π restricts to a finite surjective generically étale morphism $U_1 \to S_1$ onto a principal open subset of $\mathbf{A}^n$, the set $W_1 := W \cap U_1$ is nonempty, $\pi|_{W_1}$ is a closed immersion, and $\pi^{-1}(\pi(W_1)) \setminus W_1 = Z' \cap U_1 = \overline{Z'} \cap U_1$ is closed in U_1. Moreover $U_1 \setminus W_1 \subseteq \Sigma$ is covered by affine open subsets on which u is regularizable, by Lemma 5.11(4). By Proposition 5.24, $u|_{U_1^{\mathrm{an}}} \in \mathcal{R}(U_1)$, so $U_1 \subseteq \Sigma$, contradicting $W_1 \neq \varnothing$. □

Corollary 5.26. *Let X be a normal projective variety over the countable field k and $L \in \mathrm{Pic}(X)_{\mathbf{Q}}$ ample. Let $\varphi\colon X^{\mathrm{an}} \to [-\infty, \infty)$ be a function, and suppose that there is a birational morphism $\mu\colon X' \to X$ from an integral projective variety X' with $\varphi \circ \mu^{\mathrm{an}} \in \mathrm{PSH}(X', \mu^* L)$. Then $\varphi \in \mathrm{PSH}(X, L)$.*

Proof. This corresponds to [Wu26, Theorem 6.5]. Replacing X' by its normalization preserves the hypothesis by pullback (Proposition 2.1(4)); we may therefore assume that X' is normal. Since μ^{an} is proper and surjective, φ is upper semicontinuous, hence bounded above. After multiplying L and φ by the same positive integer, we may assume that L is a very ample line bundle, with sections $z_0, \ldots, z_N$ defining an embedding. Put $U_i := X_{z_i}$. Then $-\log|z_i| = \max_j \log|z_j/z_i|$ on U_i^{an}, and this function lies in $\mathrm{FS}(U_i)$. By Lemma 5.4(2), applied on X' to the semiample line bundle $\mu^* L$, the function $u_i := \varphi - \log|z_i|$ is weakly psh on U_i^{an} with witness μ, and it has logarithmic growth.

Let $Z \subsetneq X$ be a closed subset such that μ is an isomorphism over $X \setminus Z$. We show that every closed point $y \in U_i \setminus Z$ has an affine open neighbourhood $W \subseteq U_i \setminus Z$ with $u_i|_{W^{\mathrm{an}}} \in \mathcal{R}(W)$. Let $y' \in X'$ be the point over y. Choose an effective Cartier divisor D on X' such that $\mathcal{O}_{X'}(D)$ is ample and $\mathrm{Supp}\, D$ contains $\mu^{-1}(Z \cup V(z_i))$ but not y'; for instance the divisor of a section, not vanishing at y', of the ideal of $\mu^{-1}(Z \cup V(z_i))$ twisted by a large multiple of an ample line bundle. Then $W' := X' \setminus \mathrm{Supp}\, D$ is an affine neighbourhood of y', which μ maps isomorphically onto an affine open neighbourhood $W \subseteq U_i \setminus Z$ of y. Let s_D be the canonical section of $\mathcal{O}_{X'}(D)$, let $H := \mu^* L \otimes \mathcal{O}_{X'}(D)$, which is ample, and $\sigma := \mu^* z_i \otimes s_D \in H^0(X', H)$, so that $X'_\sigma = W'$. By Proposition 2.1(3),(5), $\Phi := \varphi \circ \mu^{\mathrm{an}} + \log|s_D|$ lies in $\mathrm{PSH}(X', H)$, and $\Phi - \log|\sigma| = u_i \circ \mu^{\mathrm{an}}$ on W'^{an}. By Lemma 5.10(1), $u_i \circ \mu^{\mathrm{an}}|_{W'^{\mathrm{an}}} \in \mathcal{R}(W')$, that is, $u_i|_{W^{\mathrm{an}}} \in \mathcal{R}(W)$. These sets W cover $U_i \setminus Z$, since they contain all its closed points.

By Theorem 5.25, $u_i \in \mathcal{R}(U_i)$. For rational $a > 1$, Lemma 5.10(2) shows that the function $\psi_{i,a}$ equal to $\varphi + (a-1)\log|z_i|$ on U_i^{an} and to $-\infty$ elsewhere lies in $\mathrm{PSH}(X, aL)$. Since $\max_i \log|z_i| = 0$ and the maximum is attained at an index i with the given point in U_i^{an}, we get $\max_i \psi_{i,a} = \varphi$. Thus $\varphi \in \mathrm{PSH}(X, aL)$ for all rational $a > 1$, and PSH2, applied to this constant family as $a \downarrow 1$, gives $\varphi \in \mathrm{PSH}(X, L)$; note that φ is finite at v_{triv} because $\varphi \circ \mu^{\mathrm{an}}$ is psh. □

6 Ample envelopes on normal varieties

In this section k is again an arbitrary algebraically closed trivially valued field. We prove the envelope property for ample classes on normal varieties. Over a countable field, it follows from the smooth case by descent along the Stein factorization of an alteration: through the birational part by Corollary 5.26, and through the finite part by Theorem 4.2. The general case is reduced to the countable one by extension of scalars.

Theorem 6.1. *Let X be a normal projective variety over k. Then $\mathrm{EP}_X(\theta)$ holds for every ample class $\theta \in N^1(X)_{\mathbf{R}}$.*

Proof. Step 1: k countable and $\theta = c_1(L)$ with $L \in \mathrm{Pic}(X)_{\mathbf{Q}}$ ample. Choose an alteration $Y \to X$ with Y smooth and projective [DJ96, Theorem 4.1], and let

$$Y \xrightarrow{\beta} T \xrightarrow{h} X$$

be its Stein factorization [Stacks, Tag 03H0]. Thus h is finite, β has geometrically connected fibres, and T is the normalization of X in the function field $k(Y)$. Since Y is normal, so is T, and since $Y \to X$ is generically finite, $k(T) = k(Y)$; hence β is birational. The variety T and the morphism β are projective, and h^*L is ample.

We prove $\mathrm{EP}_T(c_1(h^*L))$ using Proposition 2.4. Let (φ_i) be a bounded-above increasing net in $\mathrm{PSH}(T, h^*L)$. The class β^*h^*L is nef, so Theorem 3.1 and Proposition 2.4 give $\Phi \in \mathrm{PSH}(Y, \beta^*h^*L)$ with

$$\Phi(w) = \sup_i \varphi_i(\beta(w)) \qquad (w \in Y^{\mathrm{div}}).$$

We claim that Φ is constant on every fibre of β^{an}. Fix $t \in T^{\mathrm{an}}$ and an admissible pair (m, s) for h^*L with $t \in T_s^{\mathrm{an}}$. By Lemma 5.4(2), applied to the semiample line bundle β^*h^*L, the function $\Phi - m^{-1}\log|\beta^*s|$ is locally psh on $\beta^{-1}(T_s)^{\mathrm{an}}$. The fibre $(\beta^{\mathrm{an}})^{-1}(t)$ is the analytification of the proper $\mathcal{H}(t)$-scheme $Y \times_T \operatorname{Spec}\mathcal{H}(t)$, which is connected since β has geometrically connected fibres; hence it is compact, connected and without boundary [Ber90, §3.4–3.5]. By Lemma 5.2(2),(4), the restriction of $\Phi - m^{-1}\log|\beta^*s|$ to it is classically psh, and $m^{-1}\log|\beta^*s|$ is constant on it. Unless Φ is identically $-\infty$ on the fibre, it attains a finite maximum there, the fibre being compact and Φ upper semicontinuous; the set where the maximum is attained is closed by upper semicontinuity and open by the local maximum principle [GR25, Theorem 6.1], hence equal to the fibre. This proves the claim.

Since β^{an} is surjective, $\Phi = \varphi \circ \beta^{\mathrm{an}}$ for a function φ on T^{an}, and Corollary 5.26 gives $\varphi \in \mathrm{PSH}(T, h^*L)$. Since β^{an} maps Y^{div} bijectively onto T^{div}, we have $\varphi = \sup_i \varphi_i$ on T^{div}, which proves $\mathrm{EP}_T(c_1(h^*L))$. Theorem 4.2 now gives $\mathrm{EP}_X(c_1(L))$.

Step 2: $\theta = c_1(L)$ with $L \in \mathrm{Pic}(X)_{\mathbf{Q}}$ ample. By Proposition 2.5(1), it suffices to show that $\mathrm{P}_L(f)$ is L-psh for every rational PL function f. After replacing L and f by a common multiple, $f = \phi_{\mathcal{L}}$ for a test configuration $(\mathcal{X}, \mathcal{L})$ for (X, L) with $\mathcal{L}$ a line bundle. Choose a finitely generated subfield of k over which X, L and $(\mathcal{X}, \mathcal{L})$ are defined, and let k_0 be its algebraic closure in k; it is countable. We obtain X_0, L_0 and $(\mathcal{X}_0, \mathcal{L}_0)$ over k_0 whose scalar extensions are X, L and $(\mathcal{X}, \mathcal{L})$. Let $r\colon X^{\mathrm{an}} \to X_0^{\mathrm{an}}$ be the natural map; then $f = f_0 \circ r$ with $f_0 \coloneqq \phi_{\mathcal{L}_0}$, since Gauss extension commutes with restriction to k_0. Normality and ampleness descend along the faithfully flat extension $k_0 \subseteq k$, so Step 1 and Proposition 2.5(1) show that $\mathrm{P}_{L_0}(f_0)$ is L_0-psh. Let $\mathfrak{a}_{m,0}$ and $\mathfrak{a}_m$ be the base ideals of $m\mathcal{L}_0$ and $m\mathcal{L}$. Flat base change [Stacks, Tag 02KH], together with the exactness of flat pullback applied to the evaluation maps, gives $\mathfrak{a}_m = \mathfrak{a}_{m,0}\mathcal{O}_{\mathcal{X}}$, hence $\phi_{\mathfrak{a}_m} = \phi_{\mathfrak{a}_{m,0}} \circ r$, and Proposition 2.5(2), applied over k_0 and over k, gives $\mathrm{P}_L(f) = \mathrm{P}_{L_0}(f_0) \circ r$. Since L_0 is ample, $\mathrm{P}_{L_0}(f_0)$ is the limit of a decreasing sequence in $\mathcal{H}^{\mathrm{gf}}_{\mathbf{Q}}(X_0, L_0)$ [BJ22, Corollary 12.18(iii) and Example 3.9]. Its pullbacks under r lie

in $\mathcal{H}^{\mathrm{gf}}_{\mathbf{Q}}(X,L)$ and decrease to the bounded function $\mathrm{P}_L(f)$, which is therefore L-psh directly by [BJ22, Definition 4.1].

Step 3: θ real. Choose rational ample classes $\theta_j \to \theta$ with $\theta_j - \theta$ ample, which is possible by openness of the ample cone and density of rational classes. Each θ_j has the envelope property by Step 2, and [BJ22, Lemma 5.9] concludes. □

7 Homogeneous division and arbitrary classes

In this section X is a normal projective variety over k. We pass from ample classes to arbitrary classes in two steps. First, the ample envelope property of Theorem 6.1 yields homogeneous Siu division (Theorem 7.2): for an effective rational Cartier divisor E, a homogeneous psh function bounded by e_E (defined below) is e_E plus a homogeneous psh function. Division transfers increasing limits of homogeneous psh functions from an ample class to an arbitrary class. Second, a psh function is recovered from its homogeneous Legendre slices, and this reduces the envelope property to increasing limits of homogeneous functions (Proposition 7.5).

7.1 Division by an effective divisor

Keep X normal and projective over k. For an effective rational Cartier divisor E, let s_E be the canonical section of $\mathcal{O}_X(mE)$ for m sufficiently divisible, and put $e_E \coloneqq m^{-1}\log|s_E|$. This is a homogeneous Fubini–Study function for $\mathcal{O}_X(E)$, hence a homogeneous psh function of class $[E]$, and $e_E(v) = -v(E)$. We first divide by tE for a fixed coefficient $t < 1$.

Lemma 7.1. *Let $\eta \in N^1(X)_{\mathbf{R}}$ and let $E \geq 0$ be a rational Cartier divisor. If $\psi \in \mathrm{PSH}_{\mathrm{hom}}(X,\eta)$ satisfies $\psi \leq e_E$, then, for every rational $0 < t < 1$, there is $q_t \in \mathrm{PSH}_{\mathrm{hom}}(X,\eta - t[E])$ such that $\psi = te_E + q_t$.*

Proof. If $E = 0$, take $q_t = \psi$. Otherwise write $E = \sum_\ell a_\ell E_\ell$ with prime divisors E_ℓ and $a_\ell > 0$. Choose a homogeneous Fubini–Study regularization $\psi_i \downarrow \psi$ with $\psi_i \in \mathcal{H}_{\mathrm{hom}}(X,L_i)$ and $c_1(L_i) \to \eta$. Since $\psi_i(\mathrm{ord}_{E_\ell}) \to \psi(\mathrm{ord}_{E_\ell}) \leq -a_\ell < -ta_\ell$, a common tail satisfies $\psi_i(\mathrm{ord}_{E_\ell}) \leq -ta_\ell$ for all ℓ. Write $\psi_i = m^{-1}\max_j \log|s_j|$ with m so divisible that mtE is Cartier. Every s_j vanishes to order at least mta_ℓ along each E_ℓ, so, X being normal, it is divisible by the canonical section of mtE. This gives $q_i \in \mathcal{H}_{\mathrm{hom}}(X, L_i - tE)$ with $\psi_i = te_E + q_i$. The q_i decrease on X^{div}, where e_E is finite, hence on X^{an} by divisorial comparison, and $q_i(v_{\mathrm{triv}}) = 0$. Since the q_i are themselves homogeneous Fubini–Study functions and their classes tend to $\eta - t[E]$, [BJ22, Definition 4.1] gives $q_t \coloneqq \lim_i q_i \in \mathrm{PSH}_{\mathrm{hom}}(X,\eta - t[E])$, and $\psi = te_E + q_t$ in the limit. □

Theorem 7.2. *Let $\theta \in N^1(X)_{\mathbf{R}}$ and let $E \geq 0$ be a rational Cartier divisor. If $\psi \in \mathrm{PSH}_{\mathrm{hom}}(X,\theta + [E])$ satisfies $\psi \leq e_E$, then $\psi = e_E + \varphi$ for some $\varphi \in \mathrm{PSH}_{\mathrm{hom}}(X,\theta)$.*

Proof. Put $u \coloneqq \psi - e_E$ on X^{div}, where both terms are finite; thus $u \leq 0$. Choose a very ample Cartier divisor A with $\theta + [A]$ ample, and let $A_1,\dots,A_r$ be the divisors of finitely many sections generating $\mathcal{O}_X(A)$. At every center one generator is a unit, so

$$\max_j e_{A_j} = 0 \qquad \text{on } X^{\mathrm{an}}. \tag{7.1}$$

Fix j and a rational $0 < \epsilon < 1$. Since $\psi + e_{A_j} \in \mathrm{PSH}_{\mathrm{hom}}(X,\theta + [A] + [E])$ is at most e_E, Lemma 7.1 gives, for rational $0 < t < 1$, functions in $\mathrm{PSH}_{\mathrm{hom}}(X,\theta + [A] + (1-t)[E])$ with divisorial values $u + e_{A_j} + (1-t)e_E$. As $t \uparrow 1$ these increase to $u + e_{A_j}$, and for t close to 1 the class $\epsilon[A] - (1-t)[E]$ is ample, so they lie in the ample class $\theta + (1+\epsilon)[A]$. By Theorem 6.1 and Proposition 2.4, there is $\chi_j \in \mathrm{PSH}_{\mathrm{hom}}(X,\theta + (1+\epsilon)[A])$ with divisorial values $u + e_{A_j}$; thus $\chi_j \leq e_{A_j}$. Lemma 7.1, applied to χ_j and A_j with coefficient $1-\epsilon$, gives $q_j \in \mathrm{PSH}_{\mathrm{hom}}(X,\theta + 2\epsilon[A])$

with divisorial values $u + \epsilon e_{A_j}$. By (7.1), $\varphi := \max_j q_j \in \mathrm{PSH}_{\mathrm{hom}}(X, \theta + 2\epsilon[A])$ has divisorial values u. By divisorial comparison φ does not depend on ϵ, and $\varphi(v_{\mathrm{triv}}) = 0$, so Lemma 2.3, applied to this constant net as $\epsilon \downarrow 0$, gives $\varphi \in \mathrm{PSH}_{\mathrm{hom}}(X, \theta)$. Finally $e_E + \varphi$ and ψ are psh and agree on X^{div}, hence everywhere. □

Remark 7.3. Theorem 7.2 is the nontrivial implication of [BJ22, Theorem 6.21], whose published proof contains a gap; this is why the coefficient t in Lemma 7.1 is kept fixed. The proof in [BJ22] takes a decreasing regularization $\psi_i \in \mathcal{H}_{\mathrm{hom}}(X, L_i)$ of $\psi \le e_E$ and rational $t_i \to 1$ with $\psi_i \le t_i e_E$, and asserts that $\psi_i' := \psi_i - t_i e_E$ decreases because ψ_i does, in order to apply [BJ22, Theorem 4.5] to the limit. For $j \ge i$, however, $\psi_j' \le \psi_i'$ amounts to $\psi_j - \psi_i \le (t_j - t_i)e_E$, which does not follow from $\psi_j \le \psi_i$ when $t_j > t_i$. For instance, let $X = \mathbf{P}^1$, E a point and $\ell := e_E$. The functions $\psi_n := (1 - \frac{1}{2n})\ell \in \mathcal{H}_{\mathrm{hom}}(X, (1 - \frac{1}{2n})E)$ decrease to ℓ, and $t_n := 1 - \frac{1}{n}$ satisfies $\psi_n \le t_n \ell$ and $t_n \to 1$, but $\psi_n' = \frac{1}{2n}\ell$ strictly increases at every divisorial point centred at E. In the proof of Theorem 7.2, divisions are only performed with a fixed coefficient, and the passage $t \uparrow 1$ is an increasing limit, handled by the ample envelope property. Our argument therefore does not use [BJ22, Theorem 6.21], nor its consequences [BJ22, Corollaries 6.22, 6.41 and Theorems 6.31, 6.35, 6.40].

Lemma 7.4. *Let X be an integral projective variety, $\theta \in N^1(X)_{\mathbf{R}}$ and $\Psi \in \mathrm{PSH}_{\mathrm{hom}}(X, \theta)$. Suppose a sequence U_n decreases pointwise and each U_n is a finite maximum*

$$U_n = \max_a (q_{n,a} + c_{n,a}), \qquad q_{n,a} \in \mathrm{PSH}_{\mathrm{hom}}(X, \theta), \quad q_{n,a} \le \Psi, \quad c_{n,a} \in \mathbf{R},$$

including the term $\Psi - n$. If $u := \lim_n U_n$ is not identically $-\infty$, then $u \in \mathrm{PSH}(X, \theta)$.

Proof. Choose decreasing homogeneous Fubini–Study approximants $p_{n,a,j} \downarrow q_{n,a}$, using the product of the finitely many index sets for each n. They can be presented in one rational line bundle $B_{n,j}$ with $c_1(B_{n,j}) \to \theta$. Indeed, choose its rational class sufficiently close to θ that its difference from each of the finitely many approximating classes is ample, and tensor with generating sections of sufficiently divisible multiples of these differences. Their homogeneous Fubini–Study functions are zero, so this changes no function. Arbitrarily small class error is compatible with arbitrarily late inner indices. Consequently

$$F_{n,j} = \max_a (p_{n,a,j} + c_{n,a}), \qquad Q_{n,j} = \max_a p_{n,a,j}$$

are respectively a Fubini–Study function and a homogeneous Fubini–Study function in $B_{n,j}$; real weights may be used in [BJ22, Definition 4.1]. Moreover, $F_{n,j} \downarrow U_n$ and $Q_{n,j} \ge \Psi$, because the carrier term is included.

For rational $0 < b < 1$ and $\epsilon > 0$, put $G(v) = F_{n,j}(bv) + \epsilon$. This is Fubini–Study in $bB_{n,j}$, and $G \ge U_n \ge u$, since the homogeneous constituents are nonpositive. We show that such functions form a directed decreasing approximation with class limit θ.

Take two old choices $G_k(v) = F_k(b_k v) + \epsilon_k$, with homogeneous parts Q_k and outer indices n_k. Choose $n \ge n_1, n_2$, rational $b > \max\{b_1, b_2\}$ and $0 < \epsilon < \min\{\epsilon_1, \epsilon_2\}$; both $1 - b$ and ϵ can be arbitrarily small. Choose rational $b_k/b < a_k < 1$. Since $q_{n,a} \le \Psi \le Q_k$, the domination property proved in Lemma 2.3 gives, for all sufficiently late inner indices,

$$p_{n,a,j} \le a_k Q_k \qquad \text{for every } a \text{ and } k = 1, 2.$$

Let $C = \max_a c_{n,a}$ and let D_k be the minimum of the constants in F_k. Then

$$F_{n,j}(bv) \le b a_k Q_k(v) + C, \qquad F_k(b_k v) \ge b_k Q_k(v) + D_k.$$

As $ba_k > b_k$, these inequalities give $G \le G_k$ on $\{Q_k < -R\}$ for sufficiently large R, independent of the remaining inner index. At $Q_k = -\infty$, domination makes all the new homogeneous constituents $-\infty$, so the comparison holds there too.

On the compact set $K_k = \{Q_k \geq -R\}$, the old function $F_k(b_k v)$ is finite. Monotonicity gives

$$U_n(bv) \leq U_{n_k}(bv) \leq U_{n_k}(b_k v) \leq F_k(b_k v).$$

The strict gap $\epsilon_k - \epsilon > 0$ and compactness therefore give $F_{n,j}(bv) + \epsilon < G_k(v)$ throughout K_k for all sufficiently late inner indices. Here the compactness argument is applied only where the upper bound is finite. Taking a common inner index for both compact sets, both tail bounds and any prescribed class accuracy proves directedness.

More precisely, order all choices by increasing n, decreasing G and decreasing $\|c_1(B_{n,j}) - \theta\| + 1 - b + \epsilon$. The construction supplies a common successor with arbitrarily small error. The resulting net decreases, its classes tend to θ, and its infimum is u: first approximate $u(v)$ by $U_n(v)$, then by $F_{n,j}(v)$, and finally let $b \uparrow 1$ and $\epsilon \downarrow 0$. For fixed Fubini–Study data, $F_{n,j}(bv) \to F_{n,j}(v)$, also at a common zero; finite thresholds handle $u(v) = -\infty$. The limit is not identically $-\infty$ by hypothesis, so [BJ22, Definition 4.1] applies. □

7.2 Envelopes from homogeneous slices

Let X be an integral projective variety over k, let $\theta \in N^1(X)_{\mathbf{R}}$, and let $u \in \mathrm{PSH}(X, \theta)$. Put $b := \sup u = u(v_{\mathrm{triv}})$. For $\lambda \in \mathbf{R}$, the *homogeneous Legendre slice* of u at λ is

$$\widehat{u}_\lambda := \inf_{t>0} \{t \cdot u - t\lambda\},$$

with the scaling action (2.1). The slices $\widehat{u}_\lambda$, $\lambda < b$, lie in $\mathrm{PSH}_{\mathrm{hom}}(X, \theta)$ without an appeal to general decreasing stability. For rational λ, apply the homogeneous projection described after Proposition 2.1 to the rational Fubini–Study approximants of $u - \lambda$. Their projections are homogeneous Fubini–Study in the same respective classes and decrease to $\widehat{u}_\lambda$, since the two infima commute; they vanish at v_{triv} because $b > \lambda$. For real $\lambda < b$, take rational parameters increasing to λ and apply Lemma 2.3. The slices decrease in λ. The one-variable convex duality calculation in [BJ22, Theorem 6.24] gives

$$u = \sup_{\lambda<b}(\widehat{u}_\lambda + \lambda). \tag{7.2}$$

This reconstruction is a pointwise identity for the convex radial profiles $t \mapsto u(tv)$; the preceding direct approximation supplies the psh assertion for its slices. Neither the endpoint continuity assertion of [BJ22, Lemma 6.23] nor [BJ22, Theorem 6.21] is used.

Proposition 7.5. *Let X be an integral projective variety and $\theta \in N^1(X)_{\mathbf{R}}$. Suppose that for every increasing net (ψ_i) in $\mathrm{PSH}_{\mathrm{hom}}(X, \theta)$ there is $\psi \in \mathrm{PSH}_{\mathrm{hom}}(X, \theta)$ with $\psi = \sup_i \psi_i$ on X^{div}. Then $\mathrm{EP}_X(\theta)$ holds.*

Proof. We verify the criterion of Proposition 2.4. Let (φ_i) be a bounded-above increasing net in $\mathrm{PSH}(X, \theta)$ and put $f := \sup_i \varphi_i$ on X^{div}. After subtracting a constant we may assume that $b_i := \sup \varphi_i$ increases to 0.

Step 1: limits of slices. Let $\lambda < 0$. For i with $b_i > \lambda$, the slices $\widehat{\varphi}_{i,\lambda}$ lie in $\mathrm{PSH}_{\mathrm{hom}}(X, \theta)$, and they increase with i, since $\varphi \mapsto \widehat{\varphi}_\lambda$ is monotone. By hypothesis there is $\psi_\lambda \in \mathrm{PSH}_{\mathrm{hom}}(X, \theta)$ with $\psi_\lambda = \sup_i \widehat{\varphi}_{i,\lambda}$ on X^{div}. The functions ψ_λ decrease in λ, because the slices do. By (7.2), after interchanging two suprema,

$$f(v) = \sup_{\lambda<0}\{\psi_\lambda(v) + \lambda\} \qquad (v \in X^{\mathrm{div}}).$$

Finally, the hypothesis applied to the increasing sequence $(\psi_{-n})_{n\geq 1}$ gives $\Psi \in \mathrm{PSH}_{\mathrm{hom}}(X, \theta)$ with $\Psi \geq \psi_\lambda$ for all $\lambda < 0$.

Step 2: a decreasing approximation of f. For $n \geq 1$, put $h_n := 2^{-n}$, let $A_n := \{-n, -n + h_n, \dots, -h_n\}$ be the set of left endpoints of the partition of $[-n, 0]$ into intervals of length h_n, and put

$$U_n := \max\Big\{\Psi - n,\ \max_{a \in A_n}(\psi_a + a + h_n)\Big\}.$$

Each U_n is θ-psh and vanishes at v_{triv}. Passing from n to $n+1$, each old interval is subdivided, which does not increase its contribution because ψ_λ decreases in λ, and the new intervals in $[-n-1, -n]$ and the new term $\Psi - n - 1$ contribute at most $\Psi - n$. Hence $U_{n+1} \leq U_n$. All its homogeneous constituents are bounded above by Ψ, and the term $\Psi - n$ is included, so Lemma 7.4 gives $U := \lim_n U_n \in \mathrm{PSH}(X, \theta)$; the limit is zero at v_{triv}. For $v \in X^{\mathrm{div}}$, the values $f(v)$ and $\Psi(v)$ are finite, and

$$f(v) \leq U_n(v) \leq \max\{\Psi(v) - n,\ f(v) + h_n\}.$$

The lower bound holds because the parameters $\lambda < -n$ contribute at most $\Psi - n$ and each $\lambda \in [a, a + h_n]$ contributes at most $\psi_a + a + h_n$; the upper bound holds because $\psi_a + a \leq f$ on X^{div}. Hence $U = f$ on X^{div}, which is the criterion of Proposition 2.4. □

Corollary 7.6. *Let X be a normal projective variety over the algebraically closed trivially valued field k. Then $\mathrm{EP}_X(\theta)$ holds for every $\theta \in N^1(X)_{\mathbf{R}}$.*

Proof. We verify the hypothesis of Proposition 7.5. Let (ψ_i) be an increasing net in $\mathrm{PSH}_{\mathrm{hom}}(X, \theta)$. Choose an effective Cartier divisor E such that $\eta := \theta + [E]$ is ample. The functions $\psi_i + e_E \in \mathrm{PSH}_{\mathrm{hom}}(X, \eta)$ increase and are bounded above by 0. By Theorem 6.1 and Proposition 2.4, there is $\chi \in \mathrm{PSH}_{\mathrm{hom}}(X, \eta)$ with $\chi = \sup_i(\psi_i + e_E)$ on X^{div}. Thus $\chi \leq e_E$ on X^{div}, hence everywhere. By Theorem 7.2, $\chi = e_E + \psi$ with $\psi \in \mathrm{PSH}_{\mathrm{hom}}(X, \theta)$, and $\psi = \sup_i \psi_i$ on X^{div}. □

8 Descent through the normalization

Let X be an integral unibranch projective variety and $\nu\colon Y := X^\nu \to X$ its normalization. We prove that psh functions descend through ν (Theorem 8.3) and deduce the main theorem. We first descend homogeneous Fubini–Study functions and then homogeneous psh functions. Reconstruction from homogeneous slices gives the general case: in positive characteristic a Frobenius power of every section upstairs descends, and in characteristic zero the additive powers of Roberts–Singh descend after a twist independent of the power.

Lemma 8.1. *The normalization ν is a finite birational universal homeomorphism, and $\nu^{\mathrm{an}}\colon Y^{\mathrm{an}} \to X^{\mathrm{an}}$ is a homeomorphism.*

Proof. The normalization is finite, varieties being Nagata [Stacks, Tag 035B and Tag 035S], and birational. It is bijective, since the points of Y over $x \in X$ correspond to the branches of $\mathcal{O}_{X,x}$ [Stacks, Tag 0C37 and Tag 0C3B], and X is unibranch [Stacks, Tag 0BQ2]. The diagonal $Y \to Y \times_X Y$ is a closed immersion, ν being finite; were it not surjective, its open complement would be a nonempty scheme of finite type over k, hence would have a k-point, that is, two distinct k-points of Y over one point of X, contradicting bijectivity. Thus ν is radicial [Stacks, Tag 01S4], hence a universal homeomorphism [Stacks, Tag 04DF]. In particular ν is bijective with purely inseparable residue field extensions. A real valuation extends uniquely through a purely inseparable extension, so ν^{an} is a continuous bijection of compact Hausdorff spaces, hence a homeomorphism. □

We henceforth identify X^{an} and Y^{an} through ν^{an}. An extension of rings $A \subseteq B$ is *subintegral* if it is integral, $\operatorname{Spec} B \to \operatorname{Spec} A$ is bijective, and the induced residue-field maps are isomorphisms. Equivalently, every finite subset of B lies in a finite tower $A \subseteq A[e_1] \subseteq \cdots \subseteq A[e_1, \dots, e_r] \subseteq B$ with $e_i^2, e_i^3 \in A[e_1, \dots, e_{i-1}]$ [Stacks, Tag 0CND]. In characteristic zero we use the additive-power criterion of Roberts–Singh in the following form.

Lemma 8.2. *Let $A \subseteq B$ be a subintegral extension of $\mathbf{Q}$-algebras and let $b \in B$. There are integers $N \geq d \geq 0$ and elements $c_1, \dots, c_d \in B$ such that*

$$a_m \coloneqq b^m + \sum_{i=1}^{\min\{d,m\}} \binom{m}{i} c_i b^{m-i} \qquad (m \geq 0) \quad \textit{satisfies} \quad a_m \in A \quad (m \geq N).$$

Moreover, whenever $N \geq d \geq 0$ and $c_1, \dots, c_d \in B$ satisfy this, $b^{n+2d} \in (a_{n+d}, \dots, a_{n+2d})B$ for every integer $n \geq 0$. This ideal-membership assertion is in B and does not require the displayed a_m to belong to A.

Proof. The first assertion follows from [RS93, Theorem 4.17, Proposition 4.2 and Corollary 4.3], after enlarging N. For the second, fix $m \geq d$, put $c_0 \coloneqq 1$ and $P(T) \coloneqq \sum_{i=0}^{d} \binom{T}{i} c_i b^{m+d-i} \in B[T]$. Then P has degree at most d, $P(m+r) = b^{d-r} a_{m+r}$ for $0 \leq r \leq d$, and $P(0) = b^{m+d}$. Lagrange interpolation over $\mathbf{Q}$ expresses $P(0)$ as a linear combination of $P(m), \dots, P(m+d)$, so $b^{m+d} \in (a_m, \dots, a_{m+d})B$; take $m = n + d$. □

Theorem 8.3. *For every $\theta \in N^1(X)_{\mathbf{R}}$, under the identification of X^{an} with Y^{an}, one has $\mathrm{PSH}(Y, \nu^*\theta) \subseteq \mathrm{PSH}(X, \theta)$.*

Proof. First let $\psi \in \mathrm{PSH}_{\mathrm{hom}}(Y, \nu^*\theta)$. By homogeneous regularization and Lemma 2.6, write $\psi_i \downarrow \psi$ with $\psi_i \in \mathcal{H}_{\mathrm{hom}}(Y, \nu^* L_i)$ and $c_1(L_i) \to \theta$, presented by sections of $\nu^*(mL_i)$ with mL_i a line bundle on X. Class promotion preserves homogeneity. By Lemma 2.3, it suffices to descend each homogeneous Fubini–Study function to a psh function in its original class. We prove this assertion in the two characteristics.

Positive characteristic $p = \operatorname{char} k$. Write $\psi = \frac{1}{m} \max_j \{\log |s_j| + \lambda_j\}$ with $s_j \in H^0(Y, \nu^*(mL))$, and fix a finite affine open cover of X trivializing mL. For U in the cover, $B_U \coloneqq \mathcal{O}_Y(\nu^{-1}U)$ is a finitely generated algebra over $A_U \coloneqq \mathcal{O}_X(U)$. Since ν is a universal homeomorphism (Lemma 8.1), each generator has a p-power in A_U by [Stacks, Tag 0CNF]; since Frobenius is a ring map, a single p-power q satisfies $b^q \in A_U$ for every $b \in B_U$ and every U. Let $f_{U,j} \in B_U$ be the coefficient of s_j in the pulled-back frame of mL over U, so that $f_{U,j}^q \in A_U$. On overlaps the coefficients change by the inverse transition function of mL, so their q-th powers change by that of qmL; as $\mathcal{O}_X \to \nu_* \mathcal{O}_Y$ is injective, the $f_{U,j}^q$ glue to $s_j^{[q]} \in H^0(X, qmL)$ with $\nu^* s_j^{[q]} = s_j^{\otimes q}$. Hence ψ coincides with $\frac{1}{qm} \max_j \{\log |s_j^{[q]}| + q\lambda_j\} \in \mathcal{H}^{\mathrm{gf}}_{\mathbf{Q}}(X, L)$, which is contained in $\mathrm{PSH}(X, c_1(L))$ by Proposition 2.1(3).

Characteristic zero. By Proposition 2.1(5), it suffices to show that for every line bundle M on X and every nonzero $s \in H^0(Y, \nu^* M)$, the function $u \coloneqq \log |s| \leq 0$ lies in $\mathrm{PSH}(X, c_1(M))$.

Choose a finite cover of X by affine opens $U_\alpha = D(\tau_\alpha)$ trivializing M, with all τ_α sections of one ample line bundle A_0, fix frames e_α of M over U_α, and write $s = f_\alpha \nu^* e_\alpha$ with $f_\alpha \in B_\alpha \coloneqq \mathcal{O}_Y(\nu^{-1}U_\alpha)$. Since $\operatorname{char} k = 0$, ν induces isomorphisms on residue fields by Lemma 8.1, so $A_\alpha \coloneqq \mathcal{O}_X(U_\alpha) \subseteq B_\alpha$ is subintegral [Stacks, Tag 0CND]. Apply Lemma 8.2 to f_α, obtaining $c_{\alpha i}$ and $a_{\alpha m}$; enlarging d (with new coefficients 0, which changes no $a_{\alpha m}$) and N, we take d and N independent of α. Put $\mathcal{Q} \coloneqq \nu_* \mathcal{O}_Y / \mathcal{O}_X$, and fix an integer R with the following properties, each of which holds for all large R: each $\tau_\alpha^R c_{\alpha i} (\nu^* e_\alpha)^{\otimes i}$ extends to a global section $\widetilde{c}_{\alpha i}$ of $\nu^*(iM + RA_0)$; multiplication by τ_α^R kills every section of a twist of $\mathcal{Q}$ by a line bundle that is supported on $V(\tau_\alpha)$ (locally, by stabilization of the kernels of multiplication by powers of τ_α on the coherent sheaf $\mathcal{Q}$); and $RA_0 + jM$ is globally generated for $0 \leq j \leq d$.

For $m \geq N$, the section

$$\widetilde{a}_{\alpha m} \coloneqq \tau_\alpha^R s^{\otimes m} + \sum_{i=1}^{d} \binom{m}{i} \widetilde{c}_{\alpha i} s^{\otimes (m-i)} \in H^0\big(Y, \nu^*(mM + RA_0)\big)$$

equals $\tau_\alpha^R a_{\alpha m} (\nu^* e_\alpha)^{\otimes m}$ over $\nu^{-1}U_\alpha$, so its image in $H^0(X, \mathcal{Q} \otimes (mM + RA_0))$ is supported on $V(\tau_\alpha)$ and killed by τ_α^R. Tensoring $0 \to \mathcal{O}_X \to \nu_*\mathcal{O}_Y \to \mathcal{Q} \to 0$ by $mM + 2RA_0$ and using left exactness of global sections, we obtain a unique $t_{\alpha m} \in H^0(X, mM + 2RA_0)$ with $\nu^* t_{\alpha m} = \tau_\alpha^R \widetilde{a}_{\alpha m}$.

Fix $n \geq \max\{1, N - d\}$. Multiplying each $t_{\alpha m}$ with $n + d \leq m \leq n + 2d$ by each member of a fixed generating system of $RA_0 + (n + 2d - m)M$ gives a finite system $\mathcal{S}_n$ of sections of $(n + 2d)M + 3RA_0$. For $v \in X^{\mathrm{an}}$, evaluating in local frames at the center, we claim

$$n\, v(s) \leq \min_{t \in \mathcal{S}_n} v(t) \leq (n + 2d)\, v(s). \tag{8.1}$$

The first inequality holds because $m - i \geq n$ in the definition of $\widetilde{a}_{\alpha m}$, so that each $\nu^* t$ with $t \in \mathcal{S}_n$ is $s^{\otimes n}$ times a global section of a line bundle on Y. For the second, choose α with $c(v) \in U_\alpha$. Then τ_α and a member of each generating system are units at $c(v)$, so $\min_t v(t) \leq v(a_{\alpha m})$ for $n + d \leq m \leq n + 2d$, and since $v \geq 0$ on B_α, Lemma 8.2 gives $\min_m v(a_{\alpha m}) \leq (n + 2d)v(f_\alpha) = (n + 2d)v(s)$. At v_{triv} both bounds vanish, so $\mathcal{S}_n$ contains a nonzero section, and the function

$$F_n \coloneqq \frac{1}{n + 2d} \max_{t \in \mathcal{S}_n} \log |t| \in \mathcal{H}_{\mathrm{hom}}\Big(X, M + \frac{3R}{n + 2d} A_0\Big)$$

satisfies $u \leq F_n \leq \frac{n}{n+2d}\, u$ by (8.1).

Choose an integer $n_0 > 2d$ with $n_0 \geq N - d$, and put $n_j \coloneqq 2^j n_0$ and $G_j \coloneqq (1 - 2d/n_j) F_{n_j}$. For $n = n_j$,

$$G_{j+1} \leq \frac{n - d}{n + d}\, u \leq \frac{n - 2d}{n}\, u \leq G_j,$$

since $u \leq 0$ and $\frac{n-d}{n+d} - \frac{n-2d}{n} = \frac{2d^2}{n(n+d)} \geq 0$. Thus G_j decreases to u pointwise, also where $u = -\infty$, its classes $(1 - 2d/n_j)(c_1(M) + \frac{3R}{n_j + 2d} c_1(A_0))$ tend to $c_1(M)$, and $G_j(v_{\mathrm{triv}}) = 0$. The G_j are homogeneous Fubini–Study functions, so [BJ22, Definition 4.1] gives $u \in \mathrm{PSH}(X, c_1(M))$ directly.

This proves reflection for homogeneous functions. Now let $\psi \in \mathrm{PSH}(Y, \nu^*\theta)$ be arbitrary and subtract its supremum, so $\sup \psi = 0$. Its slices $\widehat{\psi}_\lambda$, $\lambda < 0$, are homogeneous psh. By Corollary 7.6 and Proposition 2.4, the increasing sequence $\widehat{\psi}_{-n}$ has a homogeneous psh upper envelope $\widetilde{\Psi}$ in the class $\nu^*\theta$, dominating every slice. Homogeneous reflection gives functions $q_\lambda, \Psi \in \mathrm{PSH}_{\mathrm{hom}}(X, \theta)$ corresponding to $\widehat{\psi}_\lambda, \widetilde{\Psi}$. The homeomorphism ν^{an} commutes with scaling, so $q_\lambda \leq \Psi$ and the Legendre reconstruction identity is unchanged. Form U_n from q_λ and Ψ using the dyadic formula in Step 2 of Proposition 7.5. These functions decrease and satisfy, pointwise under the identification of the analytifications,

$$\psi \leq U_n \leq \max\{\Psi - n, \psi + 2^{-n}\}.$$

Since $\Psi \leq 0$, the right-hand side tends to ψ, including where $\psi = -\infty$. Thus $U_n \downarrow \psi$, and Lemma 7.4 proves $\psi \in \mathrm{PSH}(X, \theta)$. Adding back its supremum finishes the proof. □

Proof of Theorem A. By Lemma 8.1, ν^{an} is a homeomorphism, so pullback commutes with pointwise suprema and usc regularization, and $\nu^* \sup^*_i \varphi_i = \sup^*_i \nu^*\varphi_i$. The functions $\nu^*\varphi_i$ lie in $\mathrm{PSH}(Y, \nu^*\theta)$ by Proposition 2.1(4). Since Y is normal, the right-hand side lies in $\mathrm{PSH}(Y, \nu^*\theta)$ by Corollary 7.6, and hence $\sup^*_i \varphi_i \in \mathrm{PSH}(X, \theta)$ by Theorem 8.3. □

Corollary 8.4. *Let X be an integral unibranch projective variety. If $\theta_i \to \theta$ and a net $u_i \in \mathrm{PSH}(X, \theta_i)$ decreases pointwise to $u \not\equiv -\infty$, then $u \in \mathrm{PSH}(X, \theta)$.*

Proof. For every v, the inequalities $u_i(v) \le u_i(v_{\mathrm{triv}})$ pass to the limit. Since $u \not\equiv -\infty$, this gives $b := u(v_{\mathrm{triv}}) = \sup u \in \mathbf{R}$. Subtract b, so that $\sup u = 0$ and $\sup u_i \ge 0$, and restrict to a tail bounded above by one fixed $u_{i_0}(v_{\mathrm{triv}})$. For each $\lambda < 0$, the homogeneous slices $\widehat{u}_{i,\lambda}$ decrease and have class θ_i. By Lemma 2.3 their limit q_λ belongs to $\mathrm{PSH}_{\mathrm{hom}}(X, \theta)$. Commuting the two infima gives

$$q_\lambda = \inf_i \inf_{t>0} \{t \cdot u_i - t\lambda\} = \inf_{t>0} \{t \cdot u - t\lambda\}.$$

The functions $t \mapsto u(tv)$ are decreasing limits of convex nonincreasing functions with a common finite upper bound, and the limit satisfies $u \le 0$. On each ray they are either finite and convex or identically $-\infty$: if the value is $-\infty$ at one positive parameter, convexity and the common upper bound force the same at every smaller parameter, and monotonicity forces it at larger parameters. For a finite radial profile, a supporting affine function at $t = 1$ has intercept at most 0, since the profile is bounded above by 0 as $t \downarrow 0$. If that intercept is 0, lower it to negative numbers tending to 0. These supporting functions give $u = \sup_{\lambda<0}(q_\lambda + \lambda)$; the identity also holds on the rays where $u = -\infty$.

By Theorem A and Proposition 2.4, the increasing homogeneous sequence q_{-n} has a homogeneous upper envelope $\Psi \in \mathrm{PSH}_{\mathrm{hom}}(X, \theta)$ dominating every q_λ. The same dyadic construction yields $U_n \downarrow u$, with constituents bounded above by Ψ and with the term $\Psi - n$. Since $u \not\equiv -\infty$, Lemma 7.4 proves the assertion. □

A The singular Yau–Tian–Donaldson conjecture

A.1 Introduction

In this appendix, we apply the envelope property to the weighted uniform Yau–Tian–Donaldson conjecture for projective varieties with klt singularities. We show that, on a klt projective variety, weighted $\mathbb{G}$-uniform K-stability is equivalent to weighted $\mathbb{G}$-coercivity of the Mabuchi functional. The proof follows the strategy of [Tru26, Appendix A], with a slight generalization to this setting. The envelope property allows us to reduce the problem on a singular variety to one on a smooth resolution.

Let X be a projective variety over $\mathbb{C}$ with at most klt singularities, and let L_X be an ample $\mathbb{Q}$-line bundle on X. Let T be an r-dimensional ($r = 0$ is allowed) real torus generated by Hamiltonian vector fields on X. Let P be the associated convex polytope (when $r = 0$, P is a single point), and let $\mathrm{Aut}_T(X)$ denote the connected subgroup of $\mathrm{Aut}(X)$ commuting with T. Let $K \subset \mathrm{Aut}_T(X)$ be a compact subgroup containing a maximal torus of $\mathrm{Aut}_T(X)$. Set $\mathbb{G} = K_{\mathbb{C}}$, and $\mathbb{T}$ as the center of $\mathbb{G}$. Fix $\mathrm{v} \in C^\infty(P, \mathbb{R}_{>0})$ and $\mathrm{w} \in C^\infty(P, \mathbb{R})$. We refer to [HL25, Section 1] for these definitions. Let $\pi\colon Y \to X$ be a $\mathbb{G}$-equivariant resolution of X, set $L_Y \coloneqq \pi^* L_X$ and $n \coloneqq \dim X$.

Theorem A.1. *With the notation above, the following statements are equivalent:*

(1) (X, L_X) *is* (v, w)*-weighted* $\mathbb{G}$*-uniformly K-stable.*

(2) The weighted Mabuchi functional $\mathbf{M}_{\mathrm{v},\mathrm{w}}$ *is* $\mathbb{G}$*-coercive.*

If, in addition, v *is log-concave,* $\mathrm{w} = \mathrm{w}_0 \cdot \mathfrak{l}$*, where* $\mathrm{w}_0 \in C^\infty(P, \mathbb{R}_{>0})$*,* $\mathfrak{l}$ *is an affine function on* P*, and the singularity on* X *is of Fano type, either statement implies the existence of a* (v, w)*-weighted cscK metric in the class* $c_1(L_X)$*.*

A.2 Preliminaries

Since X is klt, it is normal, and Y is smooth. The envelope property for (X, L_X) and for (Y, L_Y), given by Corollary 7.6, together with [HL25, Proposition 2.10, Lemma 3.4], gives the following proposition.

Proposition A.2. *With the notation of Appendix A.1, let* $(X^{\mathrm{an}}, L_X^{\mathrm{an}})$ *and* $(Y^{\mathrm{an}}, L_Y^{\mathrm{an}})$ *be the Berkovich analytifications of* (X, L_X) *and* (Y, L_Y)*, respectively. Then*

(1) Pullback by π *is a bijection from the* K*-invariant functions in* $\mathrm{PSH}(X^{\mathrm{an}}, L_X^{\mathrm{an}})$ *onto the* K*-invariant functions in* $\mathrm{PSH}(Y^{\mathrm{an}}, L_Y^{\mathrm{an}})$*, and any* K*-invariant non-Archimedean Fubini–Study metric on* $(Y^{\mathrm{an}}, L_Y^{\mathrm{an}})$ *is the pullback of a* K*-invariant non-Archimedean Fubini–Study metric on* $(X^{\mathrm{an}}, L_X^{\mathrm{an}})$*.*

(2) For any normal model $(\mathcal{Y}, \mathcal{L}_Y)$ *of* (Y, L_Y)*, there exists a normal model* $(\mathcal{X}, \mathcal{L}_X)$ *of* (X, L_X) *such that* $P_{L_Y}(\phi_{\mathcal{L}_Y}) = \pi^* P_{L_X}(\phi_{\mathcal{L}_X})$*.*

(3) For any big normal model $(\mathcal{Y}, \mathcal{L}_Y)$ *of* (Y, L_Y)*, let* $\mathfrak{a}_m$ *be the base ideal of* $|m\mathcal{L}_Y|$*. Then* $\phi_m = \phi_{\mathcal{L}_Y} + \frac{1}{m}\varphi_{\mathfrak{a}_m}$ *converges uniformly to* $P_{L_Y}(\phi_{\mathcal{L}_Y})$*.*

(4) For any $\mathbb{G}$*-equivariant normal model* $(\mathcal{X}, \mathcal{L}_X)$*, we have*

$$\pi_* \mathrm{MA}_{\mathrm{v}}\Big(\pi^* P_{L_X}(\phi_{\mathcal{L}_X})\Big) = \mathrm{MA}_{\mathrm{v}}\Big(P_{L_X}(\phi_{\mathcal{L}_X})\Big),$$

where the weighted non-Archimedean Monge–Ampère operator MA_{v} *is defined in [HL25, Definition 3.1].*

Remark A.3. Passing to a higher model, or replacing $\mathcal{L}_Y$ by $\mathcal{L}_Y + c\,\mathcal{Y}_0$ (which shifts $P_{L_Y}(\phi_{\mathcal{L}_Y})$ by the constant c), changes neither $\mathbf{M}^{\mathrm{NA}}_{\mathrm{v,w}}$ nor $\mathbf{J}^{\mathrm{NA}}_{\mathrm{v},\mathbb{T}}$. Hence uniform K-stability for models may be tested on $\mathbb{G}$-equivariant big SNC models.

We also use the following positive-intersection formula for the non-Archimedean entropy from [HL25, Proposition 4.1], which is based on [Li23; Li22].

Proposition A.4. *Let $(\mathcal{Y}, \mathcal{L}_Y)$ be a $\mathbb{G}$-equivariant big SNC model of (Y, L_Y) and $\phi \coloneqq P_{L_Y}(\phi_{\mathcal{L}_Y})$. Then*

$$\mathbf{H}^{\mathrm{NA}}_{\mathrm{v}}(\phi) = \langle {\mathcal{L}_Y}^n \rangle_{\mathrm{v}} \cdot K^{\log}_{\mathcal{Y}/Y_{\mathbb{P}^1}},$$

where the weighted positive intersection $\langle \mathcal{L}_Y^n \rangle_{\mathrm{v}} \cdot D$ with a divisor D on $\mathcal{Y}$ is defined in [HL25, Proposition 4.1].

By [HL25, Proposition 4.3, Corollary 4.4], weighted $\mathbb{G}$-uniform K-stability lifts from (X, L_X) to (Y, L_Y) as follows.

Proposition A.5. *With the notation of Appendix A.1, (X, L_X) is (v, w)-weighted $\mathbb{G}$-uniformly K-stable for models if and only if (Y, L_Y) is (v, w)-weighted $\mathbb{G}$-uniformly K-stable for models.*

Proof of Theorem A.1. Assume (1), and let $(\mathcal{Y}, \mathcal{L}_Y)$ be a $\mathbb{G}$-equivariant big SNC model of (Y, L_Y). For sufficiently divisible m, let $\mathfrak{a}_m$ be the base ideal of $|m\mathcal{L}_Y|$ and $\phi_m \coloneqq \phi_{\mathcal{L}_Y} + \frac{1}{m}\varphi_{\mathfrak{a}_m}$. By Proposition A.2(1), ϕ_m is the pullback of a K-invariant Fubini–Study metric on $(X^{\mathrm{an}}, L_X^{\mathrm{an}})$, attached to a $\mathbb{G}$-equivariant test configuration of (X, L_X) with the same values of $\mathbf{M}^{\mathrm{NA}}_{\mathrm{v,w}}$ and $\mathbf{J}^{\mathrm{NA}}_{\mathrm{v},\mathbb{T}}$ [HL25, Proposition 4.3]. Hence (1) gives $\gamma > 0$, independent of m and of the model, such that

$$\mathbf{M}^{\mathrm{NA}}_{\mathrm{v,w}}(\phi_m) \geq \gamma\, \mathbf{J}^{\mathrm{NA}}_{\mathrm{v},\mathbb{T}}(\phi_m). \tag{A.1}$$

By Proposition A.2(3), ϕ_m converges uniformly, hence strongly, to $\phi = P_{L_Y}(\phi_{\mathcal{L}_Y})$. Thus $\mathbf{R}^{\mathrm{NA}}_{\mathrm{v}}$ and $\mathbf{E}^{\mathrm{NA}}_{\mathrm{w}}$ converge along (ϕ_m), $\mathbf{J}^{\mathrm{NA}}_{\mathrm{v},\mathbb{T}}$ converges by [HL25, Lemma 3.10], and $\mathbf{H}^{\mathrm{NA}}_{\mathrm{v}}$ converges by Proposition A.6. Letting $m \to \infty$ in (A.1) gives $\mathbf{M}^{\mathrm{NA}}_{\mathrm{v,w}}(\phi) \geq \gamma\, \mathbf{J}^{\mathrm{NA}}_{\mathrm{v},\mathbb{T}}(\phi)$. By Remark A.3, (Y, L_Y) is (v, w)-weighted $\mathbb{G}$-uniformly K-stable for models, hence so is (X, L_X) by Proposition A.5, and [HL25, Theorem 1.2] gives (2).

Conversely, assume (2): there are $\gamma, C > 0$ such that $\mathbf{M}_{\mathrm{v,w}}(\Phi_t) \geq \gamma\, \mathbf{J}_{\mathrm{v},\mathbb{T}}(\Phi_t) - C$ for every K-invariant geodesic ray Φ in $\mathcal{E}^1_K(X, L_X)$. For every $\mathbb{G}$-equivariant test configuration of (X, L_X), with Fubini–Study metric ϕ, the slope formulas [HL25, Lemma 3.9, Corollary 3.31] along the associated K-invariant geodesic ray give $\mathbf{M}^{\mathrm{NA}}_{\mathrm{v,w}}(\phi) \geq \gamma\, \mathbf{J}^{\mathrm{NA}}_{\mathrm{v},\mathbb{T}}(\phi)$, which is (1).

Finally, under the additional hypotheses, (2) and [PT26, Theorem A] give a singular (v, w)-weighted cscK metric in $c_1(L_X)$; the proof there combines the uniform coercivity estimate of [BJT26, Theorem A] on the resolution with uniform a priori estimates. □

A.3 Entropy approximation

Let $(\mathcal{Y}, \mathcal{L}_Y)$ be a $\mathbb{G}$-equivariant big SNC model of (Y, L_Y) over $\mathbb{P}^1$, and let $\mathfrak{a}_m$ be the base ideal of $|m\mathcal{L}_Y|$. Set $\phi_m = \phi_{\mathcal{L}_Y} + \frac{1}{m}\varphi_{\mathfrak{a}_m}$ and $\phi = P_{L_Y}(\phi_{\mathcal{L}_Y})$.

Proposition A.6. *With the notation above, we have*

$$\lim_{m\to\infty} \mathbf{H}^{\mathrm{NA}}_{\mathrm{v}}(\phi_m) = \mathbf{H}^{\mathrm{NA}}_{\mathrm{v}}(\phi). \tag{A.2}$$

Proof. There exist a $\mathbb{G}$-equivariant projective birational morphism $\rho\colon \mathcal{Y}' \to \mathcal{Y}$ with $\mathcal{Y}'$ smooth, an ample $\mathbb{Q}$-divisor A and an effective $\mathbb{Q}$-divisor G on $\mathcal{Y}'$, and an effective $\mathbb{Q}$-divisor D in the class of $\rho^*\mathcal{L}_Y$ such that

$$D \sim_{\mathbb{Q}} A + G,$$

and $\operatorname{Supp} G = \mathbf{B}_+(\rho^*\mathcal{L}_Y)$. Unlike in [Tru26, Appendix A], G may have horizontal components.

Let m be a sufficiently divisible positive integer. Set $\mathfrak{a}'_m = \mathfrak{a}_m \cdot \mathcal{O}_{\mathcal{Y}'}$, and let $p_m \colon \mathcal{Y}_m \to \mathcal{Y}'$ be the normalized blowup of $\mathfrak{a}'_m$. Let $r_m \colon \tilde{\mathcal{Y}}_m \to \mathcal{Y}'$ be a $\mathbb{G}$-equivariant log resolution of $\mathfrak{a}'_m$, which factors through $q_m \colon \tilde{\mathcal{Y}}_m \to \mathcal{Y}_m$. On $\mathcal{Y}_m$, there exist a semiample divisor P_m and an effective divisor F_m such that

$$p_m^*(mD) = P_m + F_m,$$

where P_m is the moving part, F_m is the fixed part. Let $M_m = \frac{1}{m} P_m$.

By Proposition A.4,

$$\mathbf{H}_{\mathrm{v}}^{\mathrm{NA}}(\phi_m) = \left(q_m^* M_m^n \cdot K^{\log}_{\tilde{\mathcal{Y}}_m/Y_{\mathbb{P}^1}}\right)_{\mathrm{v}}, \qquad \mathbf{H}_{\mathrm{v}}^{\mathrm{NA}}(\phi) = \langle \mathcal{L}_Y^n \rangle_{\mathrm{v}} \cdot K^{\log}_{\mathcal{Y}/Y_{\mathbb{P}^1}}.$$

On the other hand, by [HL25, Proposition 4.1],

$$\lim_{m\to\infty} \left(q_m^* M_m^n \cdot K^{\log}_{\mathcal{Y}/Y_{\mathbb{P}^1}}\right)_{\mathrm{v}} = \langle \mathcal{L}_Y^n \rangle_{\mathrm{v}} \cdot K^{\log}_{\mathcal{Y}/Y_{\mathbb{P}^1}}.$$

Moreover

$$0 \le \left(q_m^* M_m^n \cdot K^{\log}_{\tilde{\mathcal{Y}}_m/Y_{\mathbb{P}^1}}\right)_{\mathrm{v}} - \left(q_m^* M_m^n \cdot K^{\log}_{\mathcal{Y}/Y_{\mathbb{P}^1}}\right)_{\mathrm{v}} \le \left(q_m^* M_m^n \cdot K_{\tilde{\mathcal{Y}}_m/\mathcal{Y}}\right)_{\mathrm{v}}.$$

To prove (A.2), it therefore suffices to show that

$$\lim_{m\to\infty} \left(q_m^* M_m^n \cdot K_{\tilde{\mathcal{Y}}_m/\mathcal{Y}}\right)_{\mathrm{v}} = 0.$$

Since v is positive and bounded, it suffices to show that

$$\lim_{m\to\infty} q_m^* M_m^n \cdot K_{\tilde{\mathcal{Y}}_m/\mathcal{Y}} = 0.$$

Since

$$\operatorname{Supp}(K_{\mathcal{Y}'/\mathcal{Y}}) \subset \operatorname{Exc}(\rho) \subset \mathbf{B}_+(\rho^*\mathcal{L}_Y),$$

we have

$$\lim_{m\to\infty} q_m^* M_m^n \cdot K_{\mathcal{Y}'/\mathcal{Y}} = \langle (\rho^*\mathcal{L}_Y)^n \rangle \cdot K_{\mathcal{Y}'/\mathcal{Y}} = 0.$$

It remains to prove that

$$\lim_{m\to\infty} q_m^* M_m^n \cdot K_{\tilde{\mathcal{Y}}_m/\mathcal{Y}'} = 0.$$

Let $\mathfrak{b}_m \coloneqq \mathcal{J}(\|\mathfrak{a}'^m_\bullet\|)$ be the asymptotic multiplier ideal on $\mathcal{Y}'$. By Nadel vanishing and Castelnuovo–Mumford regularity, there are integers N and m_0, with N dividing m_0 and ND, $m_0 A$, $m_0 G$ integral and $m_0 D \sim m_0 A + m_0 G$, such that $\mathcal{O}_{\mathcal{Y}'}(mD + m_0 A) \otimes \mathfrak{b}_m$ is globally generated for all $m \ge 1$ divisible by N; below m runs through such integers. Since $(m+m_0)D \sim mD + m_0 A + m_0 G$, this gives $\mathfrak{b}_m \cdot \mathcal{O}_{\mathcal{Y}'}(-m_0 G) \subseteq \mathfrak{a}'_{m+m_0}$. For $m > m_0$, as $\mathfrak{b}_m \subseteq \mathfrak{b}_{m-m_0}$ and $\mathfrak{a}'_m \subseteq \mathcal{J}(\mathfrak{a}'_m) \subseteq \mathfrak{b}_m$, we get $\mathfrak{b}_m \cdot \mathcal{O}_{\mathcal{Y}'}(-m_0 G) \subseteq \mathfrak{a}'_m \subseteq \mathfrak{b}_m$, hence

$$0 \le \operatorname{ord}_B(\mathfrak{a}'_m) - \operatorname{ord}_B(\mathfrak{b}_m) \le m_0 \operatorname{ord}_B(G)$$

for every prime divisor B over $\mathcal{Y}'$.

Let E be a prime exceptional divisor of $p_m \colon \mathcal{Y}_m \to \mathcal{Y}'$. By [Tru26, Proposition A.1],

$$0 \le \operatorname{ord}_E(K_{\mathcal{Y}_m/\mathcal{Y}'}) \le (n+1)\Big(\operatorname{ord}_E(\mathfrak{a}'_m) - \operatorname{ord}_E(\mathcal{J}(\mathfrak{a}'_m))\Big),$$

where the first inequality follows from the fact that $\mathcal{Y}'$ is smooth. Since $\mathcal{J}(\mathfrak{a}'_m) \subseteq \mathfrak{b}_m$, we have

$$0 \le \operatorname{ord}_E(K_{\mathcal{Y}_m/\mathcal{Y}'}) \le (n+1)\Big(\operatorname{ord}_E(\mathfrak{a}'_m) - \operatorname{ord}_E(\mathfrak{b}_m)\Big) \le (n+1) m_0 \operatorname{ord}_E(G).$$

By the projection formula,

$$q_m^* M_m^n \cdot K_{\tilde{\mathcal{Y}}_m/\mathcal{Y}'} = M_m^n \cdot K_{\mathcal{Y}_m/\mathcal{Y}'}.$$

Applying the preceding inequality and [HL25, Proposition 4.1] again, we obtain

$$0 \leq \overline{\lim_{m\to\infty}}\, q_m^* M_m^n \cdot K_{\tilde{\mathcal{Y}}_m/\mathcal{Y}'} \leq (n+1)m_0 \lim_{m\to\infty} M_m^n \cdot G = (n+1)m_0 \langle (\rho^* \mathcal{L}_Y)^n \rangle \cdot G = 0,$$

where the last equality follows from $\operatorname{Supp} G = \mathbf{B}_+(\rho^* \mathcal{L}_Y)$. This completes the proof. □

References

[AC] N. Bourbaki. Algèbre commutative. Springer Science & Business Media, 2007.

[BE21] S. Boucksom and D. Eriksson. Spaces of norms, determinant of cohomology and Fekete points in non-Archimedean geometry. *Adv. Math.* 378 (2021), Paper No. 107501, 124. URL: https://doi.org/10.1016/j.aim.2020.107501.

[Ber90] V. G. Berkovich. Spectral theory and analytic geometry over non-Archimedean fields. Vol. 33. Mathematical Surveys and Monographs. Providence, RI: American Mathematical Society, 1990.

[BJ22] S. Boucksom and M. Jonsson. Global pluripotential theory over a trivially valued field. *Annales de la Faculté des Sciences de Toulouse. Mathématiques* 31.3 (2022), pp. 647–836. arXiv: 1801.08229 [math.AG]. URL: https://doi.org/10.5802/afst.1705.

[BJ24] S. Boucksom and M. Jonsson. Addendum to the article "Global pluripotential theory over a trivially valued field". *Annales de la Faculté des Sciences de Toulouse. Mathématiques* 33.2 (2024), pp. 405–417. arXiv: 2206.07183 [math.AG]. URL: https://doi.org/10.5802/afst.1775.

[BJT26] S. Boucksom, M. Jonsson, and A. Trusiani. Weighted extremal Kähler metrics on resolutions of singularities. *Comm. Pure Appl. Math.* 79.4 (2026), pp. 1073–1148. URL: https://doi.org/10.1002/cpa.70026.

[BMS08] M. Blickle, M. Mustaţă, and K. E. Smith. Discreteness and rationality of F-thresholds. *Michigan Mathematical Journal* 57 (2008). Special volume in honor of Melvin Hochster, pp. 43–61. URL: https://doi.org/10.1307/mmj/1220879396.

[Can20] E. Canton. Berkovich log discrepancies in positive characteristic. *Pure and Applied Mathematics Quarterly* 16.5 (2020), pp. 1465–1532. arXiv: 1711.03002 [math.AG]. URL: https://doi.org/10.4310/PAMQ.2020.v16.n5.a5.

[CLD25] A. Chambert-Loir and A. Ducros. Formes différentielles réelles et courants sur les espaces de Berkovich. Version 2. July 25, 2025. arXiv: 1204.6277v2 [math.AG]. URL: https://arxiv.org/abs/1204.6277v2 (visited on 09/26/2026).

[DJ96] A. J. de Jong. Smoothness, semi-stability and alterations. *Publications Mathématiques de l'IHÉS* 83 (1996), pp. 51–93.

[FN80] J. E. Fornæss and R. Narasimhan. The Levi problem on complex spaces with singularities. *Math. Ann.* 248.1 (1980), pp. 47–72. URL: https://doi.org/10.1007/BF01349254.

[GJKM19] W. Gubler, P. Jell, K. Künnemann, and F. Martin. Continuity of plurisubharmonic envelopes in non-archimedean geometry and test ideals. *Ann. Inst. Fourier (Grenoble)* 69.5 (2019). With an appendix by José Ignacio Burgos Gil and Martín Sombra, pp. 2331–2376. URL: https://doi.org/10.5802/aif.3296.

[GR25] W. Gubler and J. Rabinoff. Classically psh and pluriharmonic functions on Berkovich spaces. Version 2. Sept. 17, 2025. arXiv: 2506.13548v2 [math.AG]. URL: https://arxiv.org/abs/2506.13548v2 (visited on 09/26/2026).

[Har77] R. Hartshorne. Algebraic Geometry. Vol. 52. Graduate Texts in Mathematics. New York-Heidelberg: Springer-Verlag, 1977, pp. xvi+496. URL: https://doi.org/10.1007/978-1-4757-3849-0.

[HL25] J. Han and Y. Liu. $\mathbb{G}$-uniform weighted K-stability for models on klt varieties. Version 2, revised 17 November 2025. 2025. arXiv: 2506.18039 [math.DG].

[HY03] N. Hara and K.-i. Yoshida. A generalization of tight closure and multiplier ideals. *Transactions of the American Mathematical Society* 355.8 (2003), pp. 3143–3174. URL: https://doi.org/10.1090/S0002-9947-03-03285-9.

[Laz04] R. Lazarsfeld. Positivity in Algebraic Geometry. I. Vol. 48. Ergebnisse der Mathematik und ihrer Grenzgebiete. Classical setting: line bundles and linear series. Berlin: Springer-Verlag, 2004, pp. xviii+387. URL: https://doi.org/10.1007/978-3-642-18808-4.

[Lel68] P. Lelong. Fonctions plurisousharmoniques et formes différentielles positives. Gordon & Breach, Paris-London-New York; distributed by Dunod Éditeur, Paris, 1968, pp. ix+79.

[Li22] C. Li. Geodesic rays and stability in the cscK problem. *Ann. Sci. Éc. Norm. Supér. (4)* 55.6 (2022), pp. 1529–1574. URL: https://doi.org/10.24033/asens.2523.

[Li23] C. Li. K-stability and Fujita approximation. *Birational Geometry, Kähler–Einstein Metrics and Degenerations*. Ed. by I. Cheltsov, X. Chen, L. Katzarkov, and J. Park. Vol. 409. Springer Proceedings in Mathematics & Statistics. Cham: Springer, 2023, pp. 545–566. arXiv: 2102.09457. URL: https://doi.org/10.1007/978-3-031-17859-7_26.

[Mus13] M. Mustaţă. The non-nef locus in positive characteristic. *A celebration of algebraic geometry*. Vol. 18. Clay Mathematics Proceedings. Providence, RI: American Mathematical Society, 2013, pp. 535–551. arXiv: 1109.3825 [math.AG].

[PT26] C.-M. Pan and T. D. Tô. Weighted cscK metrics on Kähler varieties. *Journal of the London Mathematical Society* 113.2 (Feb. 2026). URL: https://doi.org/10.1112/jlms.70442.

[RS93] L. G. Roberts and B. Singh. Subintegrality, invertible modules and the Picard group. *Compositio Mathematica* 85.3 (1993), pp. 249–279.

[ST12] K. Schwede and K. Tucker. A survey of test ideals. *Progress in commutative algebra 2*. Berlin: Walter de Gruyter, 2012, pp. 39–99.

[Stacks] The Stacks Project Authors. Stacks Project. http://stacks.math.columbia.edu. 2020.

[Swa80] R. G. Swan. On seminormality. *Journal of Algebra* 67.1 (1980), pp. 210–229.

[Tru26] A. Trusiani. A solution to the Yau–Tian–Donaldson Conjecture through Special Fujita Approximations. Version 2, revised 17 August 2026; Appendix A by Sébastien Boucksom. 2026. arXiv: 2605.30063.

[Wu26] L. Wu. On the local theory of non-archimedean psh functions. Version 1. Sept. 23, 2026. arXiv: 2609.27610v1 [math.AG]. URL: https://arxiv.org/abs/2609.27610v1 (visited on 09/26/2026).

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